



Burning bridges or bridging divides: Geopolitics and collaboration in research and innovation[☆]

Shreyas Gadgin Matha^a, Matte Hartog^a, Christian Chacua^a, Pierre-Alexandre Balland^{a,b}, Ricardo Hausmann^a, Muhammed Ali Yıldırım^{a,c,d,*}

^a The Growth Lab, Harvard Kennedy School, Cambridge, MA, United States

^b Centre for European Policy Studies, Brussels 1000, Belgium

^c Global Linkages Lab, Brown University, Providence, RI, USA

^d Department of Economics, Koç University, Sarıyer, Istanbul, Turkey

ARTICLE INFO

JEL classification:

F50

F60

O30

Keywords:

Research and innovation

Collaborations

Geopolitics

Globalization

Technological sovereignty

ABSTRACT

This paper analyzes the association between geopolitical alignment and international research and innovation collaborations using data spanning 2000–2023. Using ideal point distance measures derived from UN General Assembly voting patterns, we find that increases in geopolitical distance are associated with reductions in international collaborations. A one-unit increase in geopolitical distance (roughly the distance between a median country pair) over three years is associated with a 3.1% reduction in collaborations in scientific publications (0.8% per standard deviation). We also find that this relationship is asymmetric: worsening geopolitical relations are associated with reductions in collaborations in scientific publications that are roughly three times larger than the gains associated with equivalent improvements, suggesting that international research partnerships are easier to damage than to build. At the technology level, critical technologies exhibit approximately three times the expected level of international collaboration compared to other fields (conditional on country-pair effects), yet are also more sensitive to shifts in geopolitical alignment, raising concerns about technology sovereignty. At the same time, despite ongoing geopolitical tensions, overall collaboration patterns have not undergone substantial changes in recent years, suggesting resilience in international research and innovation networks.

1. Introduction

Since World War II, globalization has transformed the movement of goods (Kolb, 2018), services (Baldwin, 2016), capital (Kalemli-Özcan and Kwak, 2020), investments (Patterson et al., 2004), and people (The World Bank, 2023), including scientists (Van Noorden, 2012) and inventors (Bahar et al., 2024). This era has fostered unprecedented number of collaborations (Hou et al., 2021) and reduced the role of distance in the exchange of ideas (Griffith et al., 2011; Li, 2014; Head et al., 2019). These developments have substantially contributed to global economic growth, scientific progress, and innovation. However, the landscape of globalization is shifting. Rising protectionism, geopolitical tensions, and fragmented supply chains threaten the free flow of goods and investment, and increasingly restrict the movement of individuals. These challenges risk hampering scientific and technological progress, particularly by fragmenting global research and innovation networks.

Despite extensive research on the determinants of inter-country collaborations (e.g., Picci, 2010), a significant gap remains in understanding how geopolitical alignment relates to collaborations. In this paper, we aim to fill this gap by examining how changes in geopolitical alignment are associated with international research and innovation collaboration patterns. Building on recent research examining political divergence and international collaborations (Charpin et al., 2024), as well as the economic costs of fragmentation (Javorcik et al., 2024; Gopinath et al., 2024; Aiyar et al., 2024; Bosone and Stamato, 2024), we use ideal point estimates from UN General Assembly voting data (Bailey et al., 2017; Voeten et al., 2009) to construct continuous measures of geopolitical distance between countries.

Using gravity models following the literature on international trade (reviewed in Head and Mayer, 2014) and innovation (Picci, 2010), we find that increases in geopolitical distance over three-year periods are negatively associated with scientific collaborations, with a 3.1% reduction per unit increase in ideal point distance (IPD) changes over

[☆] This article is part of a Special issue entitled: 'Rethinking science & technology' published in Research Policy.

* Corresponding author at: The Growth Lab, Harvard Kennedy School, Cambridge, MA, United States.

E-mail address: muhammed.yildirim@hks.harvard.edu (M.A. Yıldırım).

a 3-year horizon. A one-unit change is roughly the same magnitude as the ideal point distance between a median country pair. The standard deviation of IPD change is approximately 0.25, implying that a one-standard-deviation increase in geopolitical divergence is associated with a 0.8% reduction in collaboration. We also find that this association is asymmetric: worsening geopolitical relations are associated with reductions in scientific collaborations that are roughly three times larger than the gains from equivalent improvements, consistent with the idea that collaborative relationships are easier to break than to build (Baldwin and Krugman, 1989; Dixit, 1989). However, despite recent geopolitical rifts, we do not observe a significant shift in overall collaboration patterns.

Global collaborations are particularly important in what we define as “critical technologies”, referring to technologies with strategic geopolitical importance. Conditional on country-pair and year effects, our estimates imply that these technologies exhibit approximately three times the number of collaborations compared to non-critical fields. We find that collaborations in these fields show a stronger association with changes in geopolitical alignment, particularly for collaborations involving the United States and China, raising important concerns about technological sovereignty and political risk.

As such, while collaboration patterns have remained relatively stable in recent years, emerging geopolitical tensions pose risks to scientific and technological partnerships. Collaborations are particularly prevalent in critical technologies, making these fields especially vulnerable. Given global challenges that rely on these critical technologies, this raises the question of how necessary science and technology can be advanced in spite of geopolitical rifts.

The paper is structured as follows. The next section provides an overview of the literature on international collaborations and presents our hypotheses. Section 3 details the data sources and outlines our methodology. In Section 4, we analyze collaboration patterns in science and technology. In Section 5, we examine the association between geopolitical distance and overall collaboration patterns, test for directional asymmetry in worsening versus improving geopolitical relations, and analyze collaborations in critical technologies. Finally, we conclude in Section 6.

2. Geopolitical distance and international collaboration

Collaborations have traditionally been predominantly local. Studies such as Jaffe et al. (1993) and Belenzon and Schankerman (2013) emphasize the critical role of geographic proximity in fostering knowledge spillovers and enabling the commercialization of technology, while McCann (2008) argues that, even in the context of globalization, the world has become more curved rather than flat over time. Local collaborations are often driven by factors such as social networks, connections to nearby institutions, and the advantages of face-to-face interactions, which build trust, improve communication, and facilitate the exchange of tacit knowledge (Breschi and Lissoni, 2001; Storper and Venables, 2004). As a result, well-known and highly specialized clusters such as Silicon Valley and the Boston–Cambridge area have emerged over the past few decades (Saxenian, 2006).

At the same time, such regions may face the risk of long-term lock-in without the introduction of external knowledge (Grabher, 1993). Trans-local collaborations can help address this limitation by bringing in new ideas and perspectives, potentially leading to more groundbreaking research over time (Adams et al., 2005). Multinational firms play a particularly important role in this process, as they can serve as conduits for cross-border knowledge transfer (Castellani et al., 2013; Keller and Yeaple, 2013; Turkina and Van Assche, 2018; Ascani et al., 2020; Crescenzi et al., 2022; Bahar et al., 2024) and act as “gatekeepers” for local clusters, enriching them with external knowledge (Morrison, 2008). Additionally, the movement of researchers represents a critical mechanism for knowledge diffusion, further supporting the exchange of ideas across regions (Ackers, 2005; Saxenian, 2006).

As a result, there is growing attention to connections that extend beyond local boundaries. To map and analyze these links, the past few decades have witnessed significant growth in the “science of science”, not only within academia but also among policymakers. Policymakers, in particular, are increasingly interested in quantifying scientific impact and leveraging these insights to enhance innovation policies and improve international competitiveness (Wagner and Jonkers, 2017; Rovito et al., 2021). International collaborations, in particular, have become a central focus. Such collaborations are crucial for large-scale projects that require resources beyond the capacity of any single country, such as the Large Hadron Collider in Europe. They can help firms enter new foreign markets (Kerr and Kerr, 2018) and facilitate the exchange of ideas and knowledge, often leading to groundbreaking discoveries and innovations. For instance, the Human Genome Project, an international effort involving scientists from around the globe, successfully mapped the human genome. Consequently, understanding how international collaborations influence a country’s scientific output and competitiveness has become an important area of interest for policymakers. The EU, for example, has emphasized the importance of international collaborations in science and innovation, actively supporting cross-border partnerships through initiatives such as Horizon 2020 (Veugeliers et al., 2015; Balland et al., 2019). These initiatives aim not only to pool resources but also to foster a globally interconnected scientific community capable of addressing complex global challenges more effectively.

Not only are advanced economies increasingly focusing on international collaborations, but emerging economies such as Brazil, China, Russia, and India are doing so as well. Reflecting this trend, a recent editorial in *Nature* (Nature Editorial, 2024, p. 258) emphasized the importance of understanding international bibliometrics for these countries, noting that “the network of emerging economic powers known as BRICS is ramping up its science collaboration. Researchers need to be involved in decisions as plans develop”.

International collaborations have experienced significant growth over the past decades. Ribeiro et al. (2018) report that the expansion of international collaboration networks has surpassed exponential rates, pointing to the emergence of a robust global innovation system. This system is characterized by a scale-free network structure that remains stable even as it expands, suggesting that international collaborations are not only increasing in scale but are also becoming more resilient and enduring over time.

However, the likelihood of collaboration is not equal across countries. Hou et al. (2021) find that countries with similar scientific and economic capacities are more likely to form closer collaborations, often shaped by factors such as shared languages, cultural backgrounds, and memberships in intergovernmental organizations. Picci (2010), in particular, finds that the likelihood of bilateral collaboration is higher when countries share a language, a common border, and cultural traits, and lower when countries are geographically distant. This suggests that international collaborations are driven not only by shared goals but also by shared identities and mutual understanding.

At the same time, geopolitical tensions affect the structure of international scientific and technological collaborations. Charpin et al. (2024) find that political divergence often results in diminished partnerships, especially in high-technology sectors where innovation and knowledge sharing are crucial. Recent studies have also documented a notable decline in productivity and international collaborations within Russian science (Zhang et al., 2024), attributed to geopolitical pressures and isolation from Western research networks. Similarly, escalating tensions between the United States and China have led to a significant reduction in collaborative initiatives, as both nations prioritize national security concerns over cooperative scientific endeavors (Shih et al., 2024). As such, our first hypothesis is as follows:

H1: Changes in geopolitical distance are negatively associated with international scientific and technological collaborations.

Beyond the average association, there are reasons to expect that this relationship is asymmetric: when two countries diverge geopolitically, the associated decline in collaborations is likely larger than the increase associated with an equivalent improvement in geopolitical alignment. International research collaborations involve substantial fixed investments in trust, complementary expertise, shared protocols, and institutional understanding (Breschi and Lissoni, 2001; Storper and Venables, 2004). When geopolitical relations worsen, these relationship-specific assets erode. When relations improve, however, new partnerships require reinvestment in these same fixed costs, and rebuilding trust and networks takes time. This asymmetry parallels hysteresis mechanisms in international trade, where sunk entry costs imply that negative shocks generate persistent effects even after conditions recover (Baldwin and Krugman, 1989; Dixit, 1989; Egger et al., 2025). Evidence from specific bilateral disruptions, in particular the EU–Russia sanctions and escalating US–China tensions, points in the same direction, with geopolitical deterioration in these cases associated with persistent declines in scientific partnerships that show no sign of reversal (Makkonen and Mitze, 2023; Aghion et al., 2024; Flynn et al., 2024). Hence:

H2: The negative association between geopolitical divergence (worsening relations) and collaborations is larger in magnitude than the positive association between geopolitical convergence (improving relations) and collaborations.

State actions structure the environment for individual and organizational behavior, shaping both macro-level geopolitical stances and micro-level research collaborations. Government-level decisions, such as visa restrictions, funding limitations, export controls, and public rhetoric, directly influence the ability and willingness of individuals and firms to engage in cross-border collaborations. These policies can impose legal, financial, and reputational risks on researchers and organizations, thereby discouraging or even preventing partnerships, irrespective of actors' inherent motivation to collaborate. Hence, while multinational corporations and mobile researchers can serve as conduits for bridging such divides (Saxenian, 2006; Qiao et al., 2024), their capacity to do so is often constrained by the broader geopolitical environment. State-level actions ultimately define the boundaries within which micro-level agency operates, shaping how easily knowledge and expertise flow across borders.

Regarding geopolitical dynamics, high-technology sectors may be particularly sensitive due to their strong links to national security and economic competitiveness. Technologies such as artificial intelligence, quantum computing, and advanced telecommunications are not only pivotal for economic growth but are also viewed as instruments of state power. Consequently, nations are increasingly cautious about sharing critical technologies with perceived rivals, fearing that such collaborations could accelerate adversarial capabilities. This, in turn, can foster a climate of mistrust and regulatory fragmentation, complicating multilateral efforts to address common challenges. It may also hamper the development of cohesive governance frameworks necessary for managing emerging technologies effectively. Hence, we propose the following hypothesis:

H3: The association between geopolitical distance and international scientific and technological collaborations is stronger for critical technologies.

Recent geopolitical events have exacerbated these trends, leading to an increase in protectionist policies. The ongoing conflict in Ukraine and intensifying US–China rivalry have created a backdrop in which countries are more inclined to prioritize domestic interests over international collaborations. This shift creates an environment in which urgent global issues, such as climate change or public health crises, may be inadequately addressed due to fractured international relations. The link, however, is not necessarily straightforward. Lee and Haupt (2021) find that collaborations between the United States and China in COVID-19 research, for instance, have actually increased. Hence, common scientific goals may sometimes transcend political rivalries, particularly in the context of urgent global challenges.

3. Methodology

To investigate the association between geopolitical distance and international collaborations, we use data on scientific publications from OpenAlex (Priem et al., 2022) and patent data from PATSTAT (European Patent Office, 2024) for the period 2000 to 2023. The geopolitical distance measures are constructed using UN General Assembly voting data from Bailey et al. (2017), which covers sessions 1 through 78, corresponding to the years 1946–2023. While recent work has shown that conclusions based on ideal point distance measures can be sensitive to specification choices (Airaudo et al., 2025), we conduct a battery of robustness checks (Section 5.3) to address this concern. A detailed list of variables, as well as sample characteristics and descriptive statistics, is provided in Appendix B. This setup is most closely related to Charpin et al. (2024), who use the FactSet database to capture industry-specific international collaborations between firms and find that political divergence particularly negatively affects high-technology collaborations. In contrast, we focus on scientific and technological collaborations, examining differences across technological domains (critical fields) and global regions over time.

3.1. Measuring international collaborations

Technology. To measure technological collaboration we rely on patent data from the Spring release of Patstat Global 2024 (European Patent Office, 2024). Patstat is maintained by the European Patent Office (EPO). The raw dataset contains bibliographical information of around 131 million patent applications, collected from multiple patent offices worldwide. However, in our analyses, we consider a processed dataset that focuses on international patent families, following Miguelez et al. (2019). Our approach groups patent applications into patent families following the DOCDB patent family definition provided by Patstat, to avoid double-counting of single inventions. In addition, we focus on foreign-oriented (international) patent families, which are “inventions for which the applicant has sought patent protection beyond its home patent office” (Miguelez et al., 2019, p. 4). This reduces the biases introduced by the over-representation of some patent offices in Patstat and by the differences in domestic patenting.

Our measures of technological collaborations are based on co-inventorship. We define a collaboration as occurring when at least two individuals are listed as inventors on a patent. To identify the countries involved in a collaboration, we use the inventors' addresses recorded at the time of patent filing. We distinguish between collaborations occurring within the boundaries of a single country (domestic collaborations) and collaborations involving inventors located in more than one country (international collaborations). Although our primary focus is on international collaborations, in some analyses we also examine changes in domestic collaborations, as these may also be related to the geopolitical environment. For example, increasing geopolitical tensions between two countries may be associated with a reconfiguration in the set of international technological partnerships and with greater protectionism in technology production.

Computationally, we define an international collaboration between two locations *A* and *B* as occurring when at least one inventor located in country *A* collaborates with at least one inventor located in country *B*. Hence, we first identify the set of distinct countries represented among the inventors on a patent, and then enumerate all unordered pairwise combinations from this set, including same-country pairs whenever at least two inventors reside in the same country. For example, if a patent has three inventors, two residing in country *A* and one in country *B*, the distinct country set is $\{A, B\}$; since country *A* has multiple inventors we retain the domestic link, yielding two relationships: *A*–*A* and *A*–*B*, each assigned equal weight. This choice is justified because we focus on the geographical extent of collaborations and seek to reduce biases that may arise from the overrepresentation

of inventors from a small number of countries and from changes in the number of inventors across locations.

Regarding weighting, we construct two types of collaboration counts. In the first, integer counts, each country pair is assigned one collaboration. As a result, the total number of collaborations equals the number of patent family–location–location combinations. In the second, fractional counts, each patent is divided by the total number of collaborations generated by that patent. Consequently, the total number of collaborations equals the number of patent families with more than one inventor.

Science. To measure scientific collaborations, we use OpenAlex database (Priem et al., 2022). This relatively new, fully open data repository, which replaces the discontinued Microsoft Academic Graph, contains metadata for over 200 million works (scientific articles, books, and related outputs), including geocoded information on the institutions with which authors are affiliated. Several studies have compared OpenAlex to traditional bibliographic databases commonly used in the literature, such as Scopus and Web of Science (Velez-Estevez et al., 2023; Akbaritabar et al., 2024; Simard et al., 2024). Most relevant to our analysis, Alperin et al. (2024) find that OpenAlex is well suited for cross-country analysis, reporting, for instance, a Spearman correlation of 0.95 in citations by country between OpenAlex and Scopus. At the same time, Simard et al. (2024) show that OpenAlex's journal indexing is more inclusive than that of Scopus and Web of Science.

In our analyses, measures of scientific collaborations are based on co-authorship. We define a collaboration as occurring when at least two individuals are listed as authors. The counting of country pairs and the assignment of weights follow the same procedure described above for patents.

Sample selection. For the gravity model analysis, we restrict the sample to countries meeting minimum thresholds for research and innovation activity. Specifically, we include only countries with at least 5000 scientific publications and at least 50 collaborative patents over the study period (2000–2023), as well as a minimum population of 1 million. These thresholds ensure that the analysis focuses on countries with sufficient research capacity and international engagement, while excluding small states and island nations with volatile collaboration patterns that could introduce noise into the estimates. In Section 5.3, we verify that the results are robust to more stringent thresholds (10 million population and 100 patents).

3.2. Geopolitical distance

We use UN voting data to measure geopolitical distance between countries. These data have been widely used since the establishment of the United Nations to study not only foreign policy behavior (Ball, 1951; Moon, 1985) but also inter-state relationships such as disputes (Gartzke, 2000) and treaty formation (Koremenos, 2005). We rely on the cross-country dataset constructed by Bailey et al. (2017), who derive ideal point estimates for countries based on their voting patterns in the United Nations General Assembly. They estimate the probability that a country votes in favor of a resolution as a function of both the resolution's content and the country's ideal point using a dynamic item-response model. These ideal points therefore measure a country's position in the international political space and serve as a proxy for geopolitical distance.

We construct three geopolitical distance measures for our empirical analysis:

Ideal Point Distance (IPD). A continuous measure, calculated as the absolute difference between countries' ideal point estimates: $IPD_{ij} = |\text{IdealPoint}_i - \text{IdealPoint}_j|$. IPD varies continuously over time as countries' voting patterns evolve.

IPD changes. Our primary measure. A dynamic continuous variable capturing shifts in geopolitical alignment, calculated as: $\Delta_h IPD_{ijt} \equiv IPD_{ijt} - IPD_{ijt-h}$ with $h \in \{1, 3, 5\}$ corresponding to 1-year, 3-year, and 5-year changes, respectively.

Different bloc indicator. A binary robustness measure that equals 1 if two countries are assigned to different geopolitical blocs and 0 if they are in the same bloc. Bloc assignments (Western, Eastern, or non-aligned) are based on UN voting patterns, are time-invariant, and rely on Gopinath et al. (2024).

Summary statistics and the correlation matrix between our key variables are given in Tables 1 and 2, respectively. Data sources for these variables can be found in Table B.1, and summary statistics for the categorical variables used in the analysis can be found in Table B.2 of Appendix.

3.3. Critical fields

We focus our analysis on critical fields such as 'Artificial Intelligence', 'Quantum technologies', and 'Solar Energy', focusing on those identified by the European Commission in their recent publications (European Commission, 2022b,a; European Commission et al., 2023). The full list of 25 fields can be found in Appendix A.¹ We apply this classification retrospectively to the entire dataset spanning 2000–2023. It is time-invariant within each technology domain; hence, a technology classified as critical remains critical throughout the analysis period, reflecting the inherent strategic importance of these technologies rather than their temporary importance as policy priorities.

To identify which patents belong to these fields, we rely on an LLM (in our case OpenAI's gpt-4o model), as fields differ substantially in content. First, for each critical field, we instruct the LLM to write four paragraphs describing the field in a nutshell. We do the same for each distinct technology code assigned to patents (646 IPC4 subclasses). We then obtain a vector embedding score for each of these descriptions, using OpenAI's most recent text embedding model at the time of writing ('text-embedding-3-large'). Subsequently, we calculate cosine similarity between the descriptions of each critical field and the patent code descriptions. For instance, four paragraphs produced by the LLM on 'Advanced sensing technologies' are very similar to those on 'Image or video recognition or understanding', thus yielding a very high score in the embedding space. For each critical technology, we present the top 10 candidate IPC4 subclasses based on semantic similarity (cosine similarity) to gpt-4o, which selects the most relevant match(es), allowing for flexible mapping where critical technologies can correspond to either a single or multiple IPC4 subclasses. Of the 641 IPC4 subclasses in our sample (out of 646), 62 (9.7%) are classified as critical technologies; these account for approximately 35% of patent families in our dataset, reflecting higher patenting activity in strategic technology domains.

3.4. Empirical strategy and estimation

To estimate bilateral collaboration flows Y_{ijt} between countries, we adopt a structural gravity framework from the international trade literature (Tinbergen, 1962; Head and Mayer, 2014; Chaney, 2018). Here, Y_{ijt} denotes the number of co-authored publications or co-invented patents for the ordered country pair i, j in year t . Although collaborations are undirected, we follow standard practice in the gravity literature (Head and Mayer, 2014; Yotov et al., 2016) and construct a directed panel in which each unordered pair appears twice (as both (i, j) and (j, i) , with identical collaboration counts). This allows the two sets

¹ This list of critical technologies was curated and kindly provided by members of the Chief Economist unit of DG Research and Innovation of the European Commission.

Table 1

Summary statistics: Key variables.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and geographic distances from CEPII.

	Mean	SD	Min	P25	Median	P75	P90	P99	Max
Scientific collaborations	667.342	2627.021	0.000	14.000	75.000	385.750	1359.000	11 352.000	71 724.000
Technological collaborations	11.922	88.154	0.000	0.000	0.000	1.564	11.000	268.675	2 260.255
Ideal point distance (lagged 1 yr)	1.076	0.830	0.000	0.335	1.002	1.611	2.150	3.335	4.610
IPD change (3 years)	-0.072	0.232	-1.568	-0.166	-0.048	0.045	0.159	0.449	1.241
Geographic distance (log)	8.487	0.933	4.007	7.876	8.759	9.186	9.465	9.811	9.886

Notes: Statistics computed using 2019 data as representative year.**Table 2**

Correlation matrix: Key variables.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and geographic distances from CEPII.

	Sci. collab.	Tech. collab.	IPD (lagged 1 yr)	IPD change (3 yr)	Distance (log)
Sci. collab.	1	.	.	.	.
Tech. collab.	0.853	1	.	.	.
IPD (lagged 1 yr)	0.009	0.030	1	.	.
IPD change (3 yr)	0.011	0.001	-0.202	1	.
Distance (log)	-0.075	-0.051	0.221	-0.093	1

Notes: Pairwise Pearson correlations between main variables used in gravity model specifications. Sci. collab. = Scientific collaborations, Tech. collab. = Technological collaborations, IPD = Ideal point distance, Distance = Geographic distance. Correlations computed using 2019 data as a representative year.

of country $\times$ year fixed effects to be separately identified as symmetric multilateral resistance proxies.

Country “masses” and multilateral resistance terms are absorbed by country $i \times$ year and country $j \times$ year fixed effects. Pair (dyad) fixed effects absorb all time-invariant bilateral heterogeneity. Identification therefore arises from within-pair changes in geopolitical alignment over time.

All specifications are estimated using Pseudo-Poisson Maximum Likelihood (PPML) with a log link to retain zero flows and obtain consistent estimates under heteroskedasticity (Santos Silva and Tenreiro, 2006). We cluster standard errors at the identifying panel level and report coefficients as percentage changes using $\% \Delta \approx 100 [\exp(\hat{\beta}) - 1]$.

We estimate five types of models:

(A) *Dyad-year changes (preferred specification).* Our preferred specification exploits within-pair changes in geopolitical alignment using h -year differences in Ideal-Point Distance (IPD) and includes pair fixed effects. Y_{ijt} denotes bilateral collaboration flows between country i and country j in year t (publications or patents); θ_{ij} are dyad (pair) fixed effects; γ_{it} and δ_{jt} are country $i \times$ year and country $j \times$ year fixed effects, respectively; and $\Delta_h \text{IPD}_{ijt} \equiv \text{IPD}_{ijt} - \text{IPD}_{ij,t-h}$ with $h \in \{1, 3, 5\}$:

$$\mathbb{E}[Y_{ijt} | \cdot] = \exp(\beta_1 \Delta_h \text{IPD}_{ijt} + \theta_{ij} + \gamma_{it} + \delta_{jt}). \quad (1)$$

Here, $\mathbb{E}[Y_{ijt} | \cdot]$ denotes the conditional mean given $\Delta_h \text{IPD}_{ijt}$, θ_{ij} , γ_{it} , and δ_{jt} , with $\Delta_h \text{IPD}_{ijt} \equiv \text{IPD}_{ijt} - \text{IPD}_{ij,t-h}$ and $h \in \{1, 3, 5\}$. We report results for $h = 3$, noting that results are qualitatively similar for $h = 1$ and $h = 5$ (Table C.3 and Table C.4 for publications and patents, respectively; Table C.2 for the technology-level analogue). The coefficient β_1 measures how collaboration responds to cumulative changes in IPD over an h -year window, rather than to the level of geopolitical distance. We prefer this change specification because it removes time-invariant confounders via pair fixed effects and focuses identification on within-pair temporal variation in geopolitical alignment. Pair fixed effects θ_{ij} absorb all time-invariant bilateral heterogeneity, while country $i \times$ year and country $j \times$ year fixed effects proxy for country masses and multilateral resistance (Head and Mayer, 2014; Piermartini and Yotov, 2016). Geography and culture controls are time-invariant and therefore collinear with θ_{ij} in this change specification. Standard errors are clustered at the dyad level.

(B) *Bloc indicator in levels (complementary specification).* We also estimate a specification with a time-invariant bloc indicator (DifferentBloc). Here, DifferentBloc $_{ij}$ equals one for pairs in different geopolitical blocs and zero otherwise:

$$\mathbb{E}[Y_{ijt} | \cdot] = \exp(\beta_2 \text{DifferentBloc}_{ij} + \gamma_{it} + \delta_{jt}). \quad (2)$$

where DifferentBloc $_{ij}$ equals one for country pairs in different geopolitical blocs, and $\mathbb{E}[Y_{ijt} | \cdot]$ denotes conditioning on DifferentBloc $_{ij}$, γ_{it} , and δ_{jt} . This specification captures associations identified from within-period variation relative to country $i \times$ year and country $j \times$ year shocks. Results appear in Table 3.

(C) *Dyad-year levels (IPD in levels; descriptive, limited interpretability).* For completeness, we also report a levels specification in which IPD enters in levels (lagged to mitigate simultaneity). Here, IPD $_{ij,t-1}$ is the lagged ideal-point distance for dyad ij :

$$\mathbb{E}[Y_{ijt} | \cdot] = \exp(\beta_1 \text{IPD}_{ij,t-1} + \gamma_{it} + \delta_{jt}). \quad (3)$$

Here, $\mathbb{E}[Y_{ijt} | \cdot]$ denotes conditioning on IPD $_{ij,t-1}$, γ_{it} , and δ_{jt} , where γ_{it} and δ_{jt} are country $i \times$ year and country $j \times$ year fixed effects. When pair fixed effects are not included, time-invariant bilateral frictions (geographic distance, common border, language, common colonizer, legal origin) can be added; they are omitted when identification rests on within-pair changes. This levels specification is descriptive and documents associations with the level of geopolitical distance but provides limited insight into how changes in geopolitical alignment relate to collaboration patterns. It can be influenced by prominent outliers (for example, the United States–China pair), which, despite large geopolitical distance, exhibit exceptionally high collaboration. Such cases can distort the regression beyond what a standard gravity setup would predict. We therefore report the IPD-in-levels results only for reference in the Appendix (Tables C.5 and C.6) and focus our analysis on the preferred IPD-in-changes specification presented above.

(D) *Technology-level analysis with critical technology interactions.* To test whether the semi-elasticity of collaboration flows with respect to changes in geopolitical distance is larger for critical technologies, we extend our preferred dyad-year changes specification to the technology level. The unit of observation becomes dyad-technology-year (i, j, k, t), where k indexes technology fields (IPC4 subclasses for patents). Our main technology-level specification includes an interaction between

Table 3

Geopolitical distance and international collaboration in publishing and patenting.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	Publications			Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
IPD change (3 yr)	−0.031*** (0.004)	−0.075*** (0.013)		−0.023* (0.010)	0.018 (0.016)	
Different bloc			−0.173*** (0.028)			0.029 (0.056)
Geographic distance (log)		−0.376*** (0.021)	−0.378*** (0.022)		−0.471*** (0.030)	−0.539*** (0.031)
Common border		−0.109+ (0.056)	−0.081 (0.058)		0.183* (0.084)	0.038 (0.089)
Common language		0.382*** (0.064)	0.395*** (0.064)		0.628*** (0.080)	0.767*** (0.082)
Common colonizer		0.339** (0.119)	0.035 (0.115)		−0.293+ (0.167)	−0.537** (0.183)
Common legal origins		0.054+ (0.033)	0.028 (0.036)		0.126* (0.056)	0.070 (0.053)
Observations	217 980	217 980	264 590	134 756	201 362	243 528
Pseudo R ²	0.994	0.956	0.956	0.964	0.945	0.944
Squared Cor.	0.997	0.898	0.891	0.995	0.953	0.952
Country $i \times$ Year FE	X	X	X	X	X	X
Country $j \times$ Year FE	X	X	X	X	X	X
Country-pair FE	X			X		

Notes: PPML estimates, where the dependent variable is the number of collaborative publications (columns 1–3) or patents (columns 4–6), between 2000–2023. Columns 1 and 4 correspond to our preferred specifications with pair fixed effects. Columns 1–2 and 4–5 report semi-elasticities with respect to IPD changes (3 yr) while columns 3 and 6 report the percentage change in collaboration when country pairs belong to different geopolitical blocs. Standard errors (in parentheses) are clustered at the dyad (country $i \times$ country j) level. All models include country $i \times$ year, country $j \times$ year, and (where indicated) country-pair fixed effects. The reduced observation count in column 4 relative to column 5 reflects the fact that nearly half of all patent country pairs have zero co-inventions across the entire sample period; these pairs are absorbed by pair fixed effects in PPML estimation. Publication collaborations (columns 1–2) are sufficiently dense that no pairs are dropped.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

IPD changes and a critical technology indicator:

$$\mathbb{E}[Y_{ijkt} | \cdot] = \exp\left(\beta_1 \Delta_3 \text{IPD}_{ijt} + \beta_2 \Delta_3 \text{IPD}_{ijt} \times \text{CriticalTech}_k + \beta_3 \text{CriticalTech}_k + \theta_{ij} + \gamma_{it} + \delta_{jt} + \zeta_k\right). \quad (4)$$

Here, Y_{ijkt} denotes bilateral collaboration flows between countries i and j in technology field k and year t ; CriticalTech_k is an indicator for critical technology fields based on the European Commission's 25 critical technology domains; ζ_k are technology fixed effects (which, when included, absorb the main effect CriticalTech_k ; in specifications without ζ_k , β_3 is separately identified); and all other notation follows the dyad-year specification above. The coefficient β_2 captures the additional semi-elasticity of collaboration flows with respect to geopolitical distance changes for critical relative to non-critical technologies. If $\beta_2 < 0$, critical technologies exhibit stronger semi-elasticities, supporting our hypothesis H3. Standard errors are clustered at the dyad-technology level (i, j, k) to account for potential correlation within technology-specific country pairs over time.

For robustness, we also estimate technology-level versions of our complementary specifications: bloc indicator models with critical technology interactions (Table C.1), and IPD-in-levels models with critical technology interactions (Table C.7). These alternative specifications help verify that our main findings are not driven by the specific choice of geopolitical measure or econometric approach.

(E) *Directional asymmetry: divergence versus convergence.* To test whether geopolitical deterioration and improvement have symmetric associations with collaboration (H2), we decompose the three-year change in IPD into its positive and negative components. Following Lieberman (1985) and York and Light (2017), who argue that directional asymmetry in social phenomena is the rule rather than the exception, we define:

$$\begin{aligned} \text{DIV}_{ijt} &= \max(\Delta_3 \text{IPD}_{ijt}, 0), \\ \text{CONV}_{ijt} &= \max(-\Delta_3 \text{IPD}_{ijt}, 0), \end{aligned} \quad (5)$$

where DIV_{ijt} captures episodes of geopolitical divergence (increasing distance) and CONV_{ijt} captures episodes of geopolitical convergence (decreasing distance). Both variables are non-negative by construction, and $\Delta_3 \text{IPD}_{ijt} = \text{DIV}_{ijt} - \text{CONV}_{ijt}$. Our asymmetry specification is:

$$\mathbb{E}[Y_{ijkt} | \cdot] = \exp\left(\beta_d \text{DIV}_{ijt} + \beta_c \text{CONV}_{ijt} + \theta_{ij} + \gamma_{it} + \delta_{jt}\right). \quad (6)$$

If the effects are symmetric, we expect $\beta_d + \beta_c = 0$ (i.e., $\beta_d < 0$ and $\beta_c = -\beta_d > 0$). We test this restriction using a Wald test. Rejection implies that the marginal association of moving toward greater geopolitical distance differs from the marginal association of moving toward alignment, that is, the “cost” of divergence exceeds the “benefit” of convergence (or vice versa). Note that this specification tests contemporaneous slope asymmetry, which is a necessary but not sufficient condition for full hysteresis, since the latter would additionally require persistence after the shock reverses. As a robustness check, we also estimate an interaction specification (Eq. (C.1) in the Appendix) that multiplies the overall IPD change by a diverging indicator ($1[\Delta_3 \text{IPD}_{ijt} > 0]$); results from this alternative specification are reported in Table C.8.

Notes on interpretation and robustness. Our analysis documents descriptive associations between geopolitical alignment and collaboration patterns. While our specifications control for many potential confounders through high-dimensional fixed effects, establishing causal relationships would require additional identifying assumptions and robustness checks beyond the scope of this study. All main specifications are estimated by PPML with high-dimensional fixed effects. We do not use log-OLS specifications given the prevalence of zeros and heteroskedasticity in collaboration data. Country $i \times$ year and country $j \times$ year fixed effects proxy for country masses and multilateral resistance, accounting for GDP, R&D, and other one-sided controls (see also Head and Mayer, 2014; Feenstra, 2015; Piermartini and Yotov, 2016). Pair fixed effects are included whenever identification relies on time-varying bilateral regressors; when pair fixed effects are absent (levels specifications), we include standard bilateral frictions to account for geography and historical ties. Note that these fixed effect specifications are highly restrictive and likely capture a lower bound of the associations.

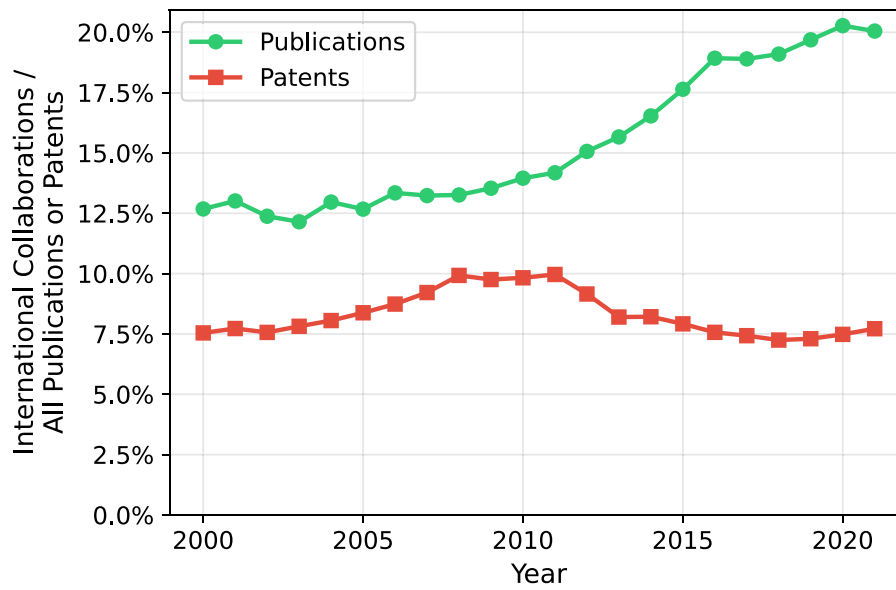


Fig. 1. Share of all publications/patents involving international collaborations.
Source: Own construction, using scientific publications from OpenAlex and patents from PATSTAT.

4. Global patterns of collaboration

We first analyze the prevalence of international collaborations. The globalization of research and innovation activities has proceeded unevenly across different domains of knowledge production. As shown in Fig. 1, international scientific publications have grown steadily from about 12.5% to 20% of total publications between 2000 and 2020, whereas international patent collaborations follow an inverted-U pattern, increasing from 7.5% to 10% until 2011 before declining to approximately 7.5% in 2020. This divergence between publications and patents suggests fundamentally different dynamics in scientific research versus technological development, with scientific knowledge production becoming increasingly globalized while invention remains primarily domestic. As patenting is inherently undertaken for protective purposes, collaboration between firms is less common, and accordingly most patent collaborations are driven by multinational companies (Bahar et al., 2024). Hence, a diverging pattern of collaborations in patenting relative to scientific publications is expected.

Countries differ significantly in their international collaboration patterns. As Fig. 2(a) shows, the United States leads in both domains, accounting for approximately 27% of all international publication collaborations and 50% of international patent collaborations. The United Kingdom accounts for 15% of international publication collaborations and 12% of international patent collaborations, while Germany accounts for 12% and 22%, respectively. China ranks fourth in international publication collaborations, with 10% of the total, but rises to third in international patent collaborations, with 13%. Japan accounts for 5% of international publication collaborations and 7% of international patent collaborations.

One of the most salient patterns in international collaboration networks is their high level of concentration. As shown in Fig. 2(b), only five international collaboration corridors, out of a total of 18,480, account for around 10% and 26% of total scientific and technological collaborations, respectively. Just 25 country pairs account for approximately 26% of all international collaborations in scientific publications and about 58% of all international collaborations in patenting.

International collaboration patterns in science and technology, measured by raw output, reflect geopolitical alignment, economic development, and institutional capacity. As shown in Figs. C.1(a) and C.1(b) in the Appendix, raw counts in both scientific publications and technological patents exhibit strong clustering along geopolitical lines,

though with distinct characteristics. While scientific collaboration networks span broadly across the Western bloc with strong connections to non-aligned countries, technological collaborations in patents display a highly concentrated pattern primarily within the Western bloc. The publications heatmap reveals the strongest collaboration intensity (shown in dark red) largely within the Western bloc, with the United States exhibiting the most extensive collaboration patterns with other Western countries, including the United Kingdom, Germany, France, Canada, and Australia. Cross-bloc collaborations are visible but substantially weaker, particularly between Western and Eastern countries, with China showing moderate collaboration levels with Western countries but limited partnerships within the Eastern bloc. Technological collaborations in patents display an even more concentrated pattern within the Western bloc, with notably weaker cross-bloc collaboration patterns, especially between Western and Eastern blocs.

In order to account for the size of collaborating countries, we calculate an enrichment score between countries to assess whether a particular country pair is over-performing or under-performing relative to the expected level of collaboration given each country's overall level of collaboration. Mathematically, if we denote the number of collaborations country i (j) engages in by N_i (N_j), the expected number of collaborations can be written as:

$$\hat{N}_{ij} = \frac{N_i N_j}{N},$$

where $N \equiv \sum_k N_k$ is the sum of all country-level collaboration counts (i.e., each collaboration is counted once for each country involved). We then calculate the enrichment score by comparing the observed number of collaborations, N_{ij} , to the expected level:

$$ES_{ij} \equiv \frac{N_{ij}}{\hat{N}_{ij}} = \frac{N_{ij}}{N_i N_j / N}. \quad (7)$$

The importance of geopolitical alignment becomes more pronounced when we account for the size of collaborating countries by comparing enrichment scores. Fig. 3(a) demonstrates that, in scientific research, collaboration largely follows geopolitical blocs, especially within the Western bloc. Additionally, smaller countries exhibit intense collaboration with their geographic and cultural neighbors, particularly among Nordic countries. To some extent, this reflects the non-linear relationship with distance, whereby small European countries located in close proximity are much more likely to collaborate with one another.

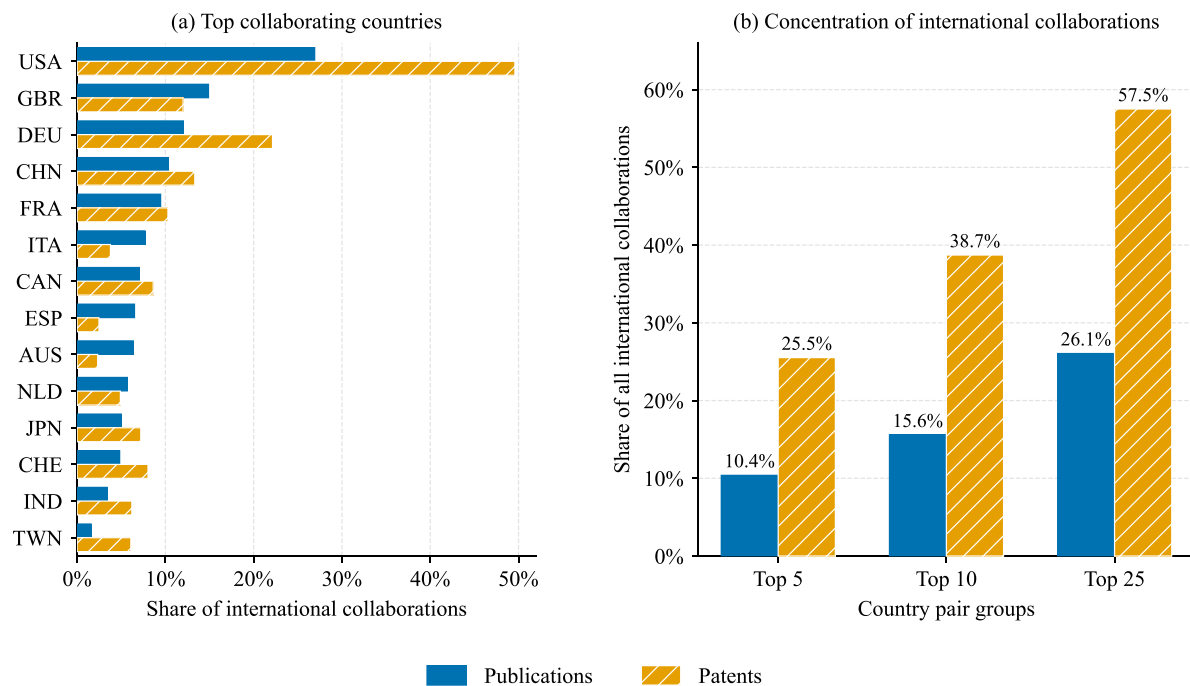


Fig. 2. Top collaborating countries and concentration of collaborations in scientific publications and patents.

Notes: (a) Countries in the top 10 for scientific or for technological collaborations (14 countries in total). Graph shows the percentage of total international collaborations accounted for by the country, sorted in descending order of collaborations in publications. (b) Here we consider only international collaborations and do not include domestic collaborations. The sample includes 225 unique countries for publications and 230 unique countries for patents, including both developed and developing economies. Total collaborating country pairs is 18,480 for publications, and 3957 for patents. Total number of international collaborations across all country pairs is 28,937,298 for publications and 575,452 for patents.

Source: Own construction, using scientific publications from OpenAlex and patents from PATSTAT.

However, as we demonstrate in the next section, geopolitical alignment remains a strong predictor of collaboration even after accounting for gravity factors. Similarly, for patents (Fig. 3(b)), we observe particularly strong intra-Western bloc collaboration patterns (shown in dark red) and notably weaker cross-bloc collaborations. The patent collaboration network also exhibits distinct regional clusters, most prominently among European countries within the Western bloc, with smaller clustering patterns also visible among Eastern bloc countries.

Fig. 4 reveals a fundamental restructuring of international scientific collaboration networks between 2000 and 2023. For publications, the most striking trend is the nearly three-fold increase (191% growth) in the share of collaborations between non-aligned countries, driven primarily by the emergence of Middle Eastern scientific hubs, particularly Saudi Arabia's partnerships with Egypt, India, and Pakistan, as well as expanding Southeast Asian regional networks such as Indonesia–Malaysia. Collaborations between Eastern and non-aligned countries grew by 70%, largely reflecting China's partnerships with Pakistan, Singapore, and India. In contrast, the share of Western–Western collaborations declined by 30% in relative terms, not because of absolute decreases but rather due to the rapid expansion of collaboration networks in previously peripheral regions. Within the Eastern–Western category, which showed modest growth of 11%, the China–USA partnership stands out as the dominant bilateral relationship, accounting for much of the growth since 2000, alongside significant increases in China's collaborations with the United Kingdom, Australia, and Canada.

The patent collaboration trends exhibit similar but more pronounced patterns. The share of Eastern–Western collaborations has stagnated since approximately 2015, with growth concentrated in the China–USA dyad (which grew by 456%) and emerging India–USA partnerships (280% growth). Meanwhile, Eastern–Eastern collaborations increased by 35%, driven by China–Russia partnerships and China's expanding ties with Central and Eastern European countries, while Eastern–non-aligned collaborations surged by 70%, reflecting China's

expansion into Singapore, India, and Saudi Arabia. These divergent trajectories suggest a gradual reorientation of innovative activity toward partnerships among emerging economies and intra-Eastern networks, potentially reflecting both geopolitical realignments and the maturation of scientific capacity in emerging economies.

Technologies with high levels of international collaboration tend to be novel or strategically important. These technology classes are therefore expected to be more vulnerable to geopolitical fragmentation. Fig. 5(a) shows the top 10 most collaborative IPC subclasses, regardless of whether they are classified as critical. The highest collaboration rates are found in biocidal pest control (19.7% of patents involve international collaboration), followed by single-substance detergent compositions (17.5%) and biocidal plant protection (17.0%). Other highly collaborative technologies include solar-powered timepieces (16.9%), heterocyclic compounds (16.6%), gravimetric detection (16.2%), animal feed production technology (15.7%), nucleic acid derivatives (15.7%), medicinal compound activity (15.2%), and oil & gas well drilling (15.2%). More broadly, the most collaborative technologies span agricultural, pharmaceutical, and medicinal technologies, specialized chemical compounds, and niche industrial applications, all significantly exceeding the mean collaboration rate of 7.6% across all technologies.

Fig. 5(b) specifically lists each critical technology and the likelihood that it involves international collaboration in invention. The figure reports international collaboration rates for each critical technology, measured as the percentage of patent families associated with that technology that involve international partnerships. Quantum technologies (16.3%) exhibit the highest rate of international collaborations, followed by geothermal energy (15.2%) and medical and pharmaceutical biotechnologies (14.1%). Beyond these, several other critical technologies—such as biofuels, advanced connectivity and navigation, industrial and environmental biotechnologies, and selected digital

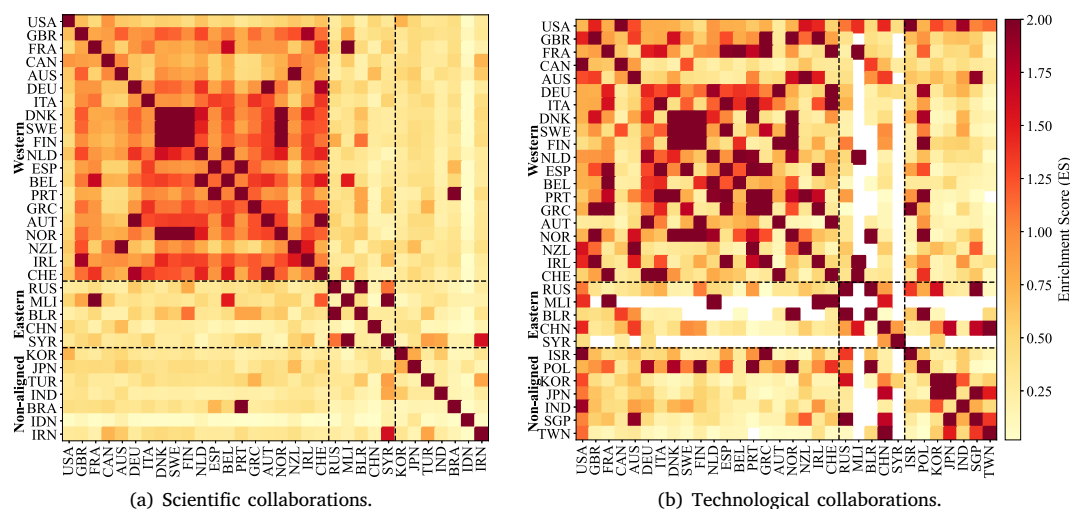


Fig. 3. Heatmap of enrichment scores in collaboration patterns.

Notes: Cell values indicate Enrichment Score (ES) of international collaborations. Enrichment Scores are calculated as in Eq. (7) using all 225 countries in the publications data and all 230 countries in the patents data. For illustrative purposes, we cap ES at the value of 2. The figure displays 20 of 22 Western and 5 of 7 Eastern bloc countries, omitting four with negligible collaboration volumes (Iceland, Luxembourg, Eritrea, and Nicaragua). For the non-aligned bloc, only the top 7 countries by collaboration volume are shown, to limit the size of the visualization. Countries are ordered by geopolitical blocs and ideal point estimates by Bailey et al. (2017). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Own construction, using scientific publications from OpenAlex and patents from PATSTAT.

technologies—also display above-average collaboration rates. In contrast, a number of fields, including advanced semiconductors, nuclear energy, robotics, battery technology, and space-related technologies, exhibit below-average collaboration, with rates typically between 4% and 7%. The mean collaboration rate across all technologies is 7.6%. In the presence of geopolitical rifts, governments may further restrict technological collaborations across countries, potentially resulting in a significant decline in innovation activity in these fields.

5. Interplay between geopolitics and collaborations

We now present our main empirical results on the relationship between geopolitical distance and international research and innovation collaborations. Our analysis focuses on the IPD changes specification with pair fixed effects (Eq. (1)), which provides the most robust identification by exploiting within-pair temporal variation in geopolitical relationships while controlling for all time-invariant bilateral characteristics. However, despite these stringent controls, we interpret this specification as descriptive rather than causal.

Table 3 presents our core findings. Increases in geopolitical distance are associated with reduced international collaborations in both scientific publications and technological patents, although the relationship differs in magnitude and robustness across domains. The results suggest that deteriorating geopolitical relationships are associated with declines in cross-border research partnerships, with implications for global knowledge flows and innovation patterns.

Our preferred specifications use three-way fixed effects (pair + country i $\times$ year + country j $\times$ year) with the 3-year change in Ideal-Point Distance (IPD). For scientific publications (Table 3, column 1), a one-unit increase in IPD over three years is associated with a 3.1% reduction in bilateral collaboration ($\exp(-0.031) - 1 \approx -3.1\%$; SE = 0.004; 95% CI: about -3.9% , -2.4%), or equivalently a 0.8% reduction per standard deviation increase in three-year changes in IPD.²

² The standard deviation of three-year changes in ideal point distance over the full period is 0.250. The per-standard-deviation change is calculated as $\exp(-0.031 \times 0.250) - 1 \approx -0.8\%$. Table 1 reports 2019-only statistics (SD = 0.232).

For patent collaborations (Table 3, column 4), the corresponding estimate is a 2.3% reduction per unit change in IPD ($\exp(-0.023) - 1 \approx -2.3\%$; SE = 0.010), statistically significant at the 5% level. Technological collaborations therefore also exhibit sensitivity to geopolitical tensions. However, this association is estimated less precisely, although the magnitude of the reduction per unit change in IPD is comparable to that observed for scientific publications. In the alternative specification with gravity controls but without pair fixed effects (column 5), the IPD-change coefficient is small and not statistically significant, suggesting that the patent response is sensitive to specification choice.

The complementary bloc specification (Table 3, columns 3 and 6) provides additional evidence using a time-invariant indicator for different geopolitical blocs. Country pairs in different blocs exhibit 15.9% lower scientific collaboration ($\exp(-0.173) - 1 \approx -15.9\%$, $p < 0.001$), while the corresponding estimate for patents is not statistically significant. This pattern is consistent with our preferred IPD changes specification and reinforces the finding that scientific collaboration is more sensitive to geopolitical alignment.

Geographic and cultural factors (Table 3, columns 2–3 and 5–6) show expected patterns. Geographic distance significantly reduces both types of collaboration, with larger effects for patents. A common language facilitates collaboration in both domains, while historical colonial relationships show mixed effects. These results are consistent with standard gravity model predictions and support the validity of our empirical approach.

We confirm the robustness of these findings through additional specifications reported in the appendix tables. Alternative measures of geopolitical distance and different time windows yield consistent results, supporting the reliability of our main estimates. The systematic pattern across specifications indicates that geopolitical tensions are associated with reduced international research collaborations, with stronger associations in scientific publications than in patents.

Explicit mass controls. While our preferred specifications absorb country masses through country i $\times$ year and country j $\times$ year fixed effects, following the structural gravity literature (Head and Mayer, 2014; Yotov et al., 2016), we also verify that our findings are robust to the explicit inclusion of mass variables, as in traditional gravity specifications (Montobbio and Sterzi, 2013). Table 4 presents specifications

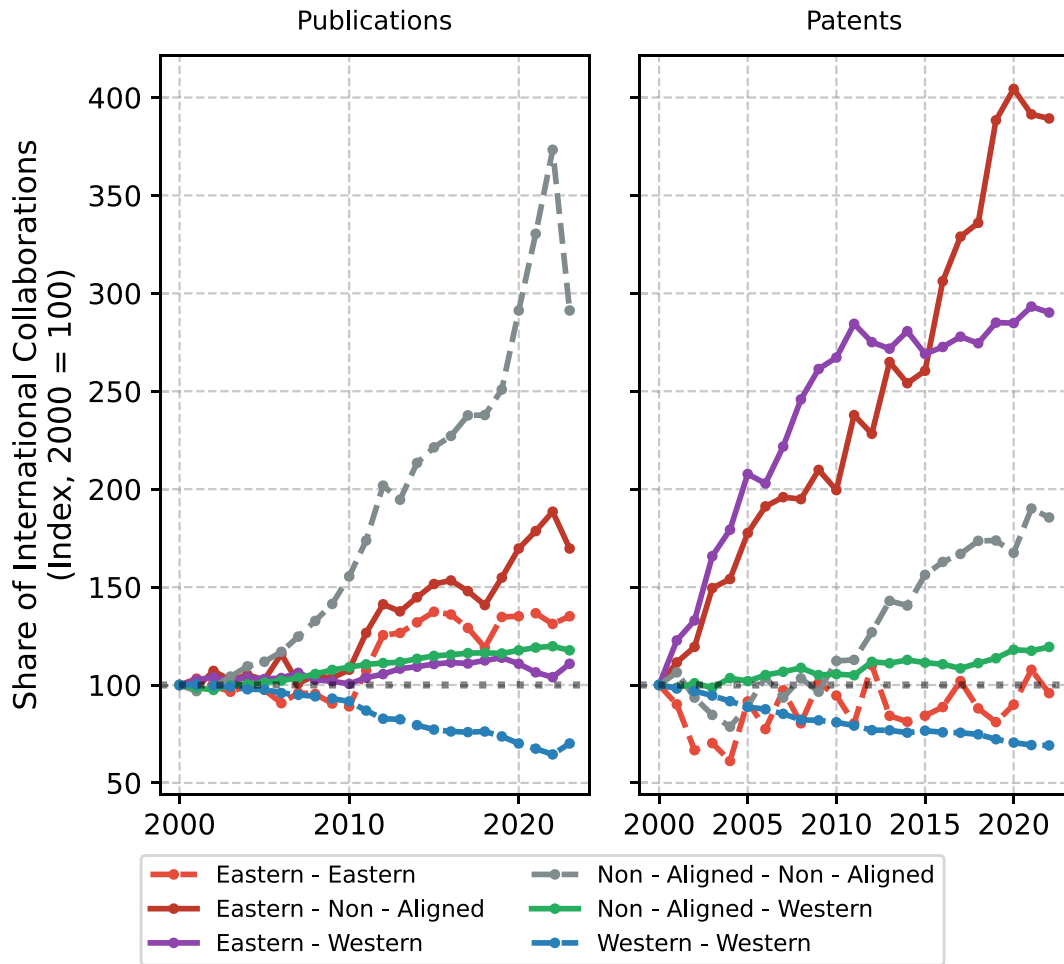


Fig. 4. Evolution of International collaborations across Blocs.

Notes: Both panels show international collaboration trends relative to year 2000 baseline (= 100). Specifically, $\left(\frac{\text{Share}_{i,j,t}}{\text{Share}_{i,j,2000}}\right) \times 100$, where $\text{Share}_{i,j,t} = \frac{\text{Collaborations}_{i,j,t}}{\text{Total Collaborations}_t}$. Lines represent different geopolitical bloc pair combinations.

Source: Own construction, using scientific publications from OpenAlex and patents from PATSTAT.

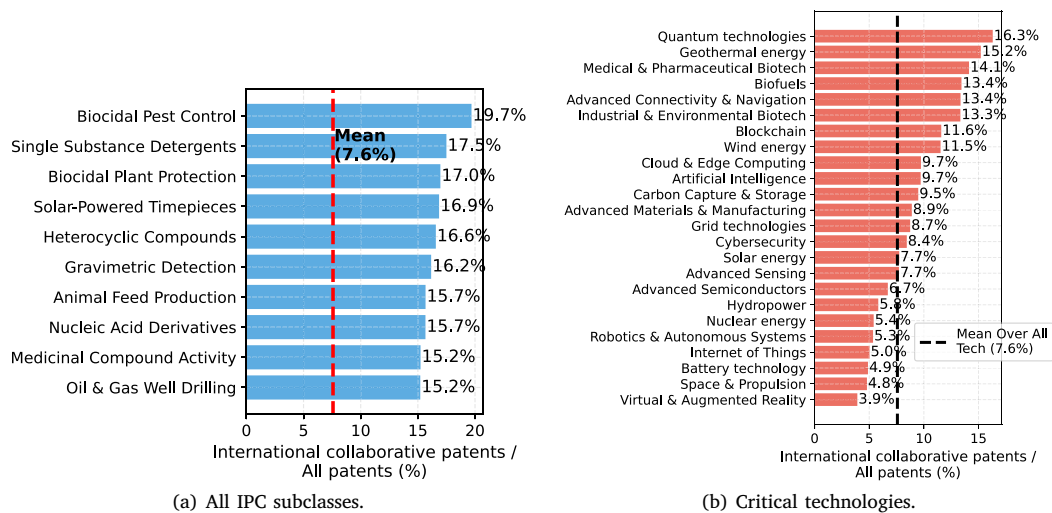


Fig. 5. Technologies by share of patent families involving international collaboration.

Notes: (a) Subclass names have been shortened for readability. The denominator (share of) is the total number of patent families in that IPC subclass. (b) The denominator (share of) is the total number of patent families across all IPC subclasses that contribute to that critical technology.

Source: Own construction using patents from PATSTAT.

Table 4

Geopolitical distance and collaboration with explicit mass controls.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, cultural and geographic distances from CEPII, and GDP from the World Bank.

	Publications			Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
IPD change (3 yr)	−0.134*** (0.023)	−0.123*** (0.023)		−0.002 (0.024)	0.015 (0.023)	
Different bloc			−0.524*** (0.045)			−0.534*** (0.117)
Geographic distance (log)	−0.576*** (0.026)	−0.481*** (0.032)	−0.377*** (0.029)	−0.650*** (0.041)	−0.446*** (0.054)	−0.299*** (0.065)
Common border	−0.441*** (0.094)	−0.464*** (0.095)	−0.402*** (0.083)	−0.103 (0.149)	−0.226 (0.156)	−0.308+ (0.173)
Common language	0.789*** (0.075)	0.770*** (0.077)	0.609*** (0.060)	1.377*** (0.102)	1.353*** (0.107)	1.151*** (0.124)
Common colonizer	−0.598*** (0.112)	−0.625*** (0.119)	−0.871*** (0.103)	−0.307 (0.220)	−0.430* (0.169)	−0.836*** (0.176)
Common legal origins	−0.088 (0.065)	−0.102 (0.065)	−0.054 (0.050)	−0.245* (0.102)	−0.281** (0.104)	−0.178+ (0.094)
Bilateral trade (log)		0.110*** (0.018)	0.121*** (0.016)		0.279*** (0.044)	0.355*** (0.047)
Country i GDP (log)	0.775*** (0.017)	0.666*** (0.027)	0.657*** (0.022)	1.084*** (0.036)	0.835*** (0.059)	0.757*** (0.056)
Country j GDP (log)	0.775*** (0.017)	0.670*** (0.025)	0.661*** (0.020)	1.084*** (0.036)	0.827*** (0.050)	0.745*** (0.048)
Observations	217 980	213 216	253 517	209 586	205 139	244 710
Pseudo R ²	0.869	0.871	0.887	0.857	0.862	0.865
Squared Cor.	0.756	0.761	0.817	0.749	0.753	0.751
Year FE	X	X	X	X	X	X

Notes: PPML estimates with explicit mass controls. Columns 1 and 4 use IPD changes (3 yr) without bilateral trade; columns 2 and 5 add bilateral trade; columns 3 and 6 use the different bloc indicator with trade. All specifications include country *i* and country *j* GDP (log) as explicit mass controls and year fixed effects rather than absorbing country masses through country $\times$ year fixed effects. Standard errors (in parentheses) are clustered at the dyad level.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

that replace the country $\times$ year fixed effects with year fixed effects and include GDP (log) from the World Bank (2024) for both countries in the dyad as explicit mass controls. Our main table (Table 3) deliberately excludes bilateral trade from the controls because trade lies on the causal path from geopolitical alignment to collaboration: geopolitical tensions reduce trade (Gopinath et al., 2024; Javorcik et al., 2024), and trade facilitates knowledge flows and collaboration. Including trade as a control would absorb part of the indirect effect of geopolitical distance, biasing the coefficient toward zero. To document this attenuation pattern, Table 4 presents both with-trade and without-trade specifications side by side.

The results are consistent with our main findings. For publications without trade controls (Table 4, column 1), the IPD change coefficient is -0.134 ($p < 0.001$), corresponding to a 12.5% reduction in collaboration per unit increase in geopolitical distance, which attenuates slightly when bilateral trade is included (column 2: -0.123 , $p < 0.001$), confirming that trade mediates part of the geopolitical association. For patents, the IPD change coefficients are small and not statistically significant (columns 4–5), consistent with the weaker patent response in our preferred specifications. The bloc specifications with trade (columns 3 and 6) show large negative associations for both domains (approximately 41% lower collaboration for cross-bloc pairs, $p < 0.001$). GDP exhibits the expected positive association with collaboration, consistent with a standard gravity relationship.

While informative, these explicit-mass specifications face important limitations. They cannot control for unobservable time-varying country-specific shocks (policy changes, institutional quality, funding priorities) that the country $\times$ year fixed effects in our preferred specifications absorb. For this reason, we continue to emphasize our preferred specifications with comprehensive fixed effects while treating the explicit-mass results as complementary evidence.

5.1. Directional asymmetry: Divergence versus convergence

Having established that geopolitical distance is negatively associated with collaboration (H1), we next test whether this relationship

Table 5

Directional asymmetry: Geopolitical distance and international collaboration. Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, and geopolitical alignment from UN General Assembly voting records.

	Publications		Patents	
	(1)	(2)	(3)	(4)
IPD change (3 yr)	−0.031*** (0.004)		−0.023* (0.010)	
IPD divergence (3 yr)		−0.047*** (0.009)		−0.008 (0.022)
IPD convergence (3 yr)		0.017* (0.008)		0.036* (0.017)
Observations	217 980	217 980	134 756	134 756
Pseudo R ²	0.994	0.994	0.964	0.964
Squared Cor.	0.997	0.997	0.995	0.995
Country $i \times$ Year FE	X	X	X	X
Country $j \times$ Year FE	X	X	X	X
Country-pair FE	X	X	X	X
Symmetry test (χ^2)		3.88		0.71
Symmetry test (<i>p</i> -value)		[0.049]		[0.400]
$ \beta_{div} / \beta_{conv} $		2.79		0.22

Notes: PPML estimates. Columns (1) and (3) replicate the preferred symmetric specification from Table 3 (columns 1 and 4) for comparison. Columns (2) and (4) decompose the 3-year IPD change into divergence ($DIV_{ijt} = \max(\Delta_3 IPD, 0)$) and convergence ($CONV_{ijt} = \max(-\Delta_3 IPD, 0)$). Standard errors (in parentheses) clustered at the dyad level. All models include country $i \times$ year, country $j \times$ year, and country-pair fixed effects.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

is symmetric, that is, whether worsening and improving geopolitical relations have equal and opposite associations with collaboration (H2). Using the decomposition specification in Eq. (6), we separate episodes of increasing geopolitical distance (divergence) from episodes of decreasing distance (convergence).

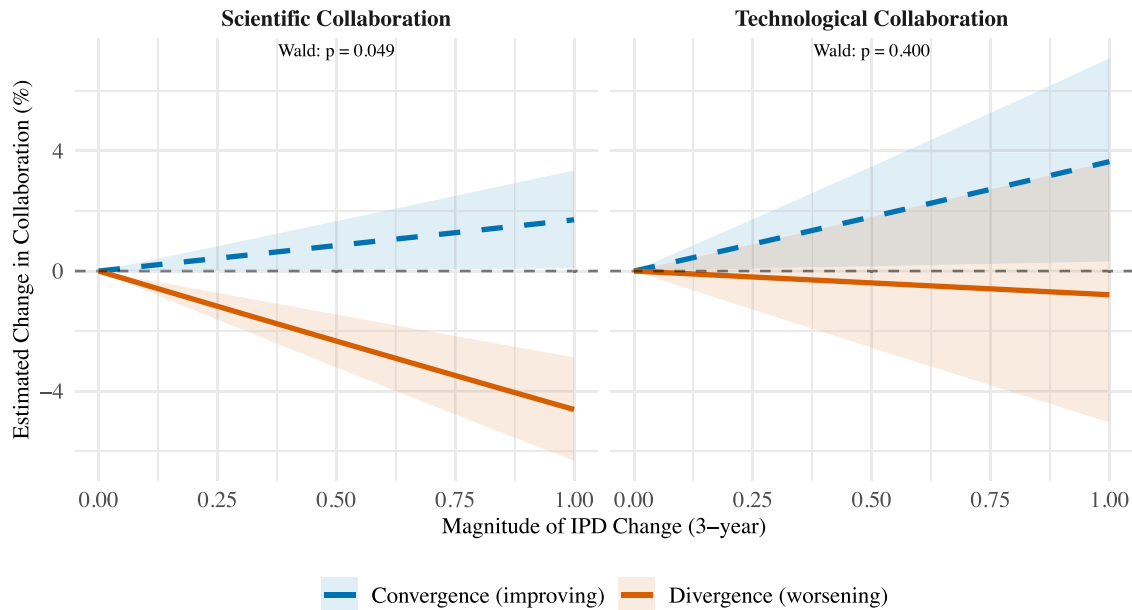


Fig. 6. Directional asymmetry: Estimated change in collaboration by direction of geopolitical shift.

Notes: Lines show the estimated percentage change in collaboration as a function of the magnitude of 3-year IPD change, separately for episodes of divergence (worsening relations, solid orange) and convergence (improving relations, dashed blue). Shaded areas represent 95% confidence intervals. Estimates are based on the decomposition specification in Eq. (6) (Table 5, columns 2 and 4). Wald test p-values test the null hypothesis of symmetry ($H_0 : \beta_d + \beta_c = 0$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, and geopolitical alignment from UN General Assembly voting records.

Table 5 presents the results. For scientific publications, the decomposition reveals a pronounced asymmetry. Worsening geopolitical relations (divergence) are associated with a 4.6% reduction in collaboration ($\hat{\beta}_d = -0.047$, $p < 0.001$), while improving relations (convergence) are associated with only a 1.7% increase ($\hat{\beta}_c = 0.017$, $p < 0.05$). The ratio of the divergence to convergence coefficients is approximately 2.8: deterioration in geopolitical relations is associated with roughly three times larger reductions in scientific collaboration than equivalent improvements. A Wald test rejects the null hypothesis of symmetry ($H_0 : \beta_d + \beta_c = 0$) at the 5% level.

For patent collaborations, the point estimates suggest a different pattern, but the results are considerably noisier. Worsening relations (divergence) show no statistically significant association ($\hat{\beta}_d = -0.008$, $p > 0.10$), while improving relations (convergence) yield a positive coefficient ($\hat{\beta}_c = 0.036$, $p < 0.05$ in the baseline). However, the convergence coefficient attains statistical significance in only two of the six sample variants examined in Table C.9, and the Wald test cannot reject symmetry ($p = 0.400$). As with the other patent collaboration results for H1, the estimates are characterized by large standard errors, making it difficult to draw firm conclusions about the directional structure of the association.

Fig. 6 visualizes this asymmetry. For scientific collaboration (left panel), the divergence line declines steeply while the convergence line rises modestly, and the Wald test rejects symmetry ($p = 0.049$). For technological collaboration (right panel), the two lines lie closer together and the confidence intervals overlap substantially; the Wald test cannot reject symmetry ($p = 0.400$).

These findings support H2 for scientific publications: worsening geopolitical relations are associated with substantially larger reductions in collaboration than the gains associated with improving relations. The asymmetry is consistent with the idea that collaborative relationships, once disrupted, may be difficult to rebuild. The sunk investments in trust, networks, and complementary expertise are lost and must be rebuilt when conditions improve. For patents, H2 is not supported; the individual coefficients are too imprecisely estimated to distinguish reliably between symmetric and asymmetric patterns, and the suggestive

point estimates are not robust across sample specifications (Table C.9). The interaction specification (Table C.8) and the sample variants in Section 5.3 confirm the robustness of the publication result.

5.2. Critical technologies

The relationship between geopolitical alignment and collaboration patterns is particularly salient for critical technologies. These technologies, which include areas such as artificial intelligence, advanced semiconductors, and quantum computing, often require substantial resources and diverse expertise, making international collaboration in innovation especially valuable. At the same time, their strategic importance for economic growth may render these technologies more sensitive to geopolitical tensions. This creates an inherent tension: the technologies that stand to benefit most from diverse expertise and international collaboration are often those most exposed to geopolitical considerations.

This tension manifests in several ways. First, critical technologies often involve significant fixed costs and specialized knowledge that may require know-how and funding from multiple countries. Second, these technologies frequently have dual-use potential, serving both civilian and military applications, which can make countries hesitant to share knowledge even as the complexity of development, and the potential for shared benefits and positive externalities, call for collaborative solutions. Third, the cumulative nature of innovation in critical technologies, particularly in general-purpose technologies, implies that restrictions on their development could have long-lasting implications for technological trajectories.

To examine whether critical technologies exhibit differential sensitivity to geopolitical distance, we extend our analysis to the technology level using patent data disaggregated by IPC classification. We estimate the technology-level specification from Eq. (4) with an interaction between IPD changes and a critical technology indicator, allowing for heterogeneous effects across technology domains. The analysis includes technology fixed effects to control for inherent differences in collaboration levels across fields.

Table 6

Geopolitical distance and international technological collaborations for critical technology.

Source: Own construction, using patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)	(5)
IPD change (3 yr) × Critical technology		−0.127** (0.040)	−0.135*** (0.022)	−0.148*** (0.021)	−0.142*** (0.038)
IPD change (3 years)	0.043*** (0.008)	0.155*** (0.036)	0.099*** (0.011)	0.075*** (0.010)	0.142*** (0.033)
Critical technology		1.078*** (0.058)			1.099*** (0.058)
Geographic distance (log)	−0.302*** (0.012)	−0.211*** (0.022)	−0.302*** (0.012)		
Common border	0.119*** (0.029)	0.077 (0.064)	0.119*** (0.029)		
Common language	0.522*** (0.031)	0.381*** (0.055)	0.521*** (0.031)		
Common colonizer	−0.253** (0.080)	−0.193 (0.124)	−0.254** (0.080)		
Common legal origins	0.056** (0.020)	0.034 (0.043)	0.056** (0.020)		
Observations	1 227 534	1 227 534	1 227 534	1 227 534	1 227 534
Pseudo R ²	0.413	0.232	0.414	0.439	0.247
Squared Cor.	0.606	0.168	0.606	0.652	0.179
Country <i>i</i> × Year FE	X	X	X	X	X
Country <i>j</i> × Year FE	X	X	X	X	X
Country-pair FE				X	X
Technology FE	X		X	X	

Notes: PPML estimates, where the dependent variable is the number of collaborative patents in each IPC subclass. Standard errors (in parentheses) are clustered at the (country *i* × country *j* × technology) level. Critical technologies are defined using the EU Commission's 25 critical technology fields. Results are qualitatively unchanged when clustering at the dyad level, with all key coefficients remaining statistically significant ($p < 0.001$).

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6 presents the results of our technology-level analysis. Columns (1) through (3) present baseline specifications with country-year fixed effects and varying controls, while columns (4) and (5) introduce country-pair fixed effects to account for time-invariant dyadic characteristics. The association with geographic distance in the baseline specifications (columns 1–3) is negative and statistically significant, suggesting that physical proximity remains important for collaboration across technological fields.

Column (5) omits technology fixed effects to allow estimation of the critical technology main effect, revealing that these strategic fields exhibit substantially higher baseline collaboration volumes. The coefficient of 1.099 indicates that the expected number of collaborative patents per dyad-year is approximately three times higher for critical technologies than for other technologies ($\exp(1.099) \approx 3.0$, or equivalently, 200% higher), conditional on country-pair and year effects. This is consistent with the above-average descriptive collaboration shares documented in Section 4, and confirms that critical technologies naturally involve more international collaboration, reflecting their complexity and resource requirements.

Most importantly, the interaction between IPD changes and critical technologies reveals a stark pattern: critical technologies are significantly more sensitive to geopolitical distance than other technologies. In our preferred specification with pair fixed effects and technology fixed effects (column 4), the interaction coefficient is -0.148 ($p < 0.001$), indicating that critical technologies face an additional 13.8% reduction in collaboration per unit increase in geopolitical distance ($\exp(-0.148) - 1 \approx -13.8\%$) beyond the baseline effect.

The main IPD-changes coefficient for non-critical technologies (column 4) is positive at 0.075 ($p < 0.001$), corresponding to a 7.8% increase ($\exp(0.075) - 1 \approx 7.8\%$). However, for critical technologies, the combined coefficient is $0.075 - 0.148 = -0.073$, implying a net 7.0% reduction ($\exp(-0.073) - 1 \approx -7.0\%$) in collaboration when geopolitical distance increases. This differential sensitivity suggests that strategic considerations are systematically associated with lower collaboration in critical technology domains relative to commercial incentives.

The positive baseline for non-critical technologies may appear to contradict the negative aggregate patent result in Table 3. However, the

two estimates are not directly comparable. The technology-level dataset includes only dyad-technology-year cells with positive collaboration, whereas the aggregate panel includes all dyad-year observations—approximately 93% of which have zero patent collaboration. The technology-level coefficient therefore captures the intensive margin (how collaboration changes among active partnerships), while the aggregate coefficient also incorporates the extensive margin (partnerships ceasing entirely). In addition, the technology fixed effects absorb cross-technology composition effects that are present in the aggregate. Indeed, when the technology-level data are summed back to dyad-year totals and re-estimated without technology fixed effects, the coefficient is negative (-0.033 , $p = 0.054$), consistent with the aggregate result. The two sets of estimates are therefore reconcilable: within surviving technology partnerships, collaboration is resilient or slightly increasing, but the overall effect is negative once the loss of entire technology collaborations is accounted for.

Further analysis reveals that this heightened sensitivity in critical technologies is concentrated among relationships involving the United States and China. As detailed in Section 5.3, when we exclude all country pairs involving either of these two powers, the differential effect for critical technologies becomes statistically insignificant. This concentration indicates that strategic considerations involving these two scientific and technological powers drive the heightened geopolitical sensitivity in critical technology collaboration.

This vulnerability has important implications for technological development paths. When geopolitical tensions are associated with reduced collaboration in critical technologies, countries may pursue parallel development efforts, potentially leading to technological divergence. This may result in incompatible standards, reduced knowledge spillovers, and duplicated research efforts. The concentration of expertise in specific technologies often follows historical patterns of specialization, implying that restrictions on collaboration may create significant barriers to entry for countries seeking to develop new capabilities in critical fields.

To verify that the differential sensitivity of critical technologies persists when using explicit mass controls rather than fixed effects, we

Table 7

Critical technology collaboration with explicit mass controls.

Source: Own construction, using patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, cultural and geographic distances from CEPII, and GDP from the World Bank.

	(1)	(2)	(3)	(4)
IPD change (3 yr) × Critical technology			−0.139*** (0.019)	−0.143*** (0.019)
IPD change (3 years)	0.030*** (0.008)	0.039*** (0.008)	0.089*** (0.008)	0.100*** (0.008)
Geographic distance (log)	−0.266*** (0.009)	−0.136*** (0.012)	−0.266*** (0.009)	−0.136*** (0.012)
Common border	0.106*** (0.028)	−0.005 (0.028)	0.106*** (0.028)	−0.006 (0.028)
Common language	0.820*** (0.026)	0.830*** (0.027)	0.820*** (0.026)	0.830*** (0.027)
Common colonizer	−0.122* (0.057)	−0.351*** (0.056)	−0.122* (0.057)	−0.352*** (0.056)
Common legal origins	−0.112*** (0.019)	−0.169*** (0.020)	−0.112*** (0.019)	−0.169*** (0.020)
Bilateral trade (log)		0.203*** (0.012)		0.203*** (0.012)
Country i GDP (log)	0.611*** (0.012)	0.448*** (0.015)	0.611*** (0.012)	0.448*** (0.015)
Country j GDP (log)	0.611*** (0.012)	0.436*** (0.014)	0.611*** (0.012)	0.436*** (0.014)
Observations	1 227 534	1 227 027	1 227 534	1 227 027
Pseudo R ²	0.347	0.354	0.347	0.354
Squared Cor.	0.475	0.485	0.476	0.486
Year FE	X	X	X	X
Technology FE	X	X	X	X

Notes: PPML estimates at the technology level with explicit mass controls. Columns (1) and (3) exclude bilateral trade; columns (2) and (4) include bilateral trade. All specifications include country *i* and country *j* GDP (log) as explicit mass controls and technology fixed effects. Standard errors (in parentheses) are clustered at the (country *i* × country *j* × technology) level.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

estimate specifications analogous to those in Table 4. Table 7 presents the corresponding technology-level analysis with GDP included as the explicit mass variable.

The results confirm the differential sensitivity of critical technologies under explicit mass controls. The interaction coefficient between IPD changes and critical technology ranges from −0.139 to −0.143 ($p < 0.001$) across specifications (columns 3–4), comparable to the −13.8% differential effect in our preferred specification (Table 6, column 4). Unlike the aggregate analysis, including trade does not attenuate the interaction coefficient. The combined effect for critical technologies remains negative across specifications, confirming the heightened geopolitical sensitivity in strategic technology domains.

These findings point to a fundamental challenge in the governance of critical technologies. The higher rate of international collaboration in these fields indicates their inherently global nature and the benefits of cross-border knowledge flows. However, the sensitivity of these collaborations to geopolitical alignment suggests that political considerations may override the technical benefits of collaboration. Restrictions on collaboration in critical technologies may lead to differing standards and protocols, divergent technological development trajectories, and, globally, a reduced pace of technological development. The concentration of expertise required for critical technologies implies that restrictions on collaboration may have particularly severe consequences for smaller countries or those with less developed technological capabilities. This presents a complex policy challenge: how to maintain the benefits of international collaboration while addressing geopolitical concerns.

5.3. Robustness: Sample composition and data construction

Beyond the robustness checks with alternative model specifications reported in Appendix C, we subject our findings to a battery of tests that vary sample composition and measurement choices to verify that the estimates are not driven by influential dyads, particular subsamples, or the construction of the collaboration measures.

We examine four dimensions. First, to assess whether these patterns are concentrated in elite networks, we restrict the sample by ex post research quality using citation thresholds (top 10% and top 50%). Second, we vary the construction of technological collaboration by replacing links based on inventors' locations with links based on applicants' locations, shifting the object of measurement from joint research activity to joint corporate filing strategies. Third, we restrict the sample to larger countries (those with a population of at least 10 million, 5000 publications, and at least 100 patents over the study period, compared to the baseline thresholds of 1 million population, 5000 publications, and 50 patents) to limit the influence of small countries with volatile collaboration patterns. Finally, we exclude all country pairs involving either the United States or China to test whether the results reflect broad patterns or are disproportionately driven by relationships involving these two countries.

Table 8 reports our preferred PPML estimates for these variants. For scientific collaborations, the IPD-change coefficients are highly stable and precisely estimated: semi-elasticities range from −2.6% (No CN/US scenario) to −3.4% (Q10 and Q50 scenarios) per one-unit increase in geopolitical distance, remaining consistently negative and statistically significant across specifications. This stability across quality filters, country selections, and measurement choices corroborates the conclusion that geopolitical distance is systematically associated with a dampening of scientific partnerships.

For technological collaborations (patents), estimates are more sensitive to sample definition but remain consistent with the main conclusion. The baseline semi-elasticity is modest (about −2.3%) and marginally significant, yet becomes materially larger when the United States and China are excluded (about −5.2%, $p < 0.01$), suggesting that the apparent resilience of patent collaboration to changes in geopolitical distance is largely driven by relationships involving these two countries. The applicant-based specification likewise yields a sizeable negative estimate (about −5.4%), indicating that corporate filing behavior does not insulate technological collaboration from geopolitical frictions.

Table 8

Sample-based robustness checks.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, and geopolitical alignment from UN General Assembly voting records.

Specification	Baseline	Q10	Q50	Applicant	No small countries	No CN/US
Publications						
IPD change (3 yr, Pair FE) ^a	−0.031*** (0.004)	−0.035*** (0.006)	−0.035*** (0.006)	−0.029*** (0.004)	−0.028*** (0.004)	−0.026*** (0.005)
Different Bloc (Controls)	−0.173*** (0.028)	−0.262*** (0.058)	−0.207*** (0.061)	−0.161*** (0.029)	−0.176*** (0.034)	−0.241*** (0.020)
Patents						
IPD change (3 yr, Pair FE) ^a	−0.023* (0.010)	−0.028 (0.019)	−0.021 (0.011)	−0.056* (0.025)	−0.022* (0.010)	−0.053** (0.018)
Different Bloc (Controls)	0.029 (0.056)	0.058 (0.054)	0.049 (0.056)	0.039 (0.085)	0.057 (0.057)	−0.217*** (0.048)
Critical technology						
IPD change × Critical (Pair FE) ^a	−0.148*** (0.021)	−0.197*** (0.032)	−0.172*** (0.026)	−0.101** (0.038)	−0.104*** (0.021)	0.034 (0.019)

Notes: PPML estimates across alternative sample and measurement specifications: (1) Baseline - standard inventor collaborations; (2) Q 10 - top 10% by citations; (3) Q 50 - top 50% by citations; (4) Applicant - corporate patent filing strategy (relevant only to patent data); (5) No small countries - countries with population of at least 10 million and at least 5000 publications and 100 patents over the study period (vs. baseline thresholds of 1 million population, 5000 publications, and 50 patents); (6) No CN/US - excluding China and the USA. “Different Bloc” uses time-invariant geopolitical classification. We show the specification type in parenthesis next to the variable names. Pair FE corresponds to the specification in columns 1 and 4 of Table 3 where we control for country $i \times$ year, country $j \times$ year, and country-pair fixed effects. With Controls, we refer to the controls in columns (2), (3), (5) and (6) of Table 3, where we directly control for the gravity variables.

^a Indicates preferred specification.

The critical technology analysis yields the most nuanced insights. Our baseline finding of heightened sensitivity to geopolitical distance in critical technologies (−13.8%) remains robust across most specifications, with particularly strong associations for high-quality patenting (top 10%: −17.9%), suggesting that the differential association between changes in geopolitical distance and collaboration in critical technologies is stronger when focusing on highly cited patents. However, when excluding China and the United States, this differential sensitivity disappears entirely, with the interaction coefficient becoming small and statistically insignificant (0.034, $\exp(0.034) - 1 \approx 3.5\%$). This pattern suggests that the heightened geopolitical sensitivity in critical technologies may be primarily driven by US–China strategic considerations rather than representing a universal phenomenon affecting all country pairs in these domains. The result also contrasts with the overall patent estimates, where excluding the United States and China makes the average IPD-change effect more negative. Taken together, the evidence indicates that US–China pairs dampen the average patent response to geopolitical distance but drive most of the critical-technology differential.

The asymmetric associations documented in Section 5.1 are similarly robust across scenarios. For publications, the divergence coefficient is negative and statistically significant ($p < 0.01$) in all six specifications, with point estimates ranging from −3.1% to −7.1% (Table C.9). The asymmetry strengthens notably when excluding China and the United States (−7.1% for divergence), consistent with the pattern observed for patents that removing these dominant dyads reveals stronger geopolitical effects. For patents, the divergence effect becomes significant only when excluding China and the United States (−9.1%, $p < 0.01$), further corroborating that these two countries exhibit atypical resilience to geopolitical deterioration in technological collaboration.

Taken together, these exercises complement the specification checks reported in Appendix C and sharpen the interpretation of the baseline results. Importantly, they demonstrate that the negative association between geopolitical distance and scientific collaboration reflects a systematic pattern across the global network rather than being driven by a small number of high-profile ruptures: the science estimates remain significant and stable in magnitude even when the most influential corridors (US–China) are removed and when the sample is restricted along multiple dimensions. For patents, excluding US and China reveals

stronger and more precisely estimated negative semi-elasticities, indicating that the baseline patent estimate is, if anything, attenuated by the resilience of US–China collaboration. In critical technologies, the global pattern of heightened sensitivity masks substantial heterogeneity, with the United States and China and their geopolitical dynamics accounting for much of the aggregate pattern.

6. Conclusion

This paper explores how geopolitical tensions relate to international research and innovation (R&I) collaborations, emphasizing the growing importance of geopolitical alignment in shaping collaboration patterns. Using ideal point distance measures derived from UN General Assembly voting patterns, our findings show that increases in geopolitical distance over three-year periods are associated with reduced collaboration. This association is asymmetric: worsening geopolitical relations are associated with roughly three times larger reductions in scientific collaboration than the gains from equivalent improvements, suggesting that preventing the breakdown of collaborative relationships may be substantially more effective than attempting to rebuild them after the fact. The negative association between geopolitical distance and collaboration is particularly pronounced in critical technologies – especially in collaborations involving the United States and China – where international partnerships are crucial. Despite these geopolitical divides, we observe a degree of stability in overall collaboration patterns, suggesting resilience in global R&I networks.

These findings suggest that geopolitical fragmentation may be associated with reduced collaborative knowledge generation, particularly in critical fields such as quantum technologies, artificial intelligence, and renewable energy. Such disruptions are concerning given that these fields are central to addressing pressing global challenges and, more broadly, to scientific and technological progress.

Additionally, the divergence between scientific and technological collaboration patterns highlights the importance of targeted interventions. The point estimates suggest greater sensitivity of scientific collaborations to geopolitical divides relative to technological partnerships, although the patent estimates are less precise and sensitive to specification choices. Technological partnerships are likely shaped by strategic and commercial considerations. Policymakers and international organizations should therefore prioritize fostering trust and cooperation, especially in critical technologies, to safeguard global innovation against the backdrop of geopolitical instability.

An important avenue for future research is to explore the role of collaborations between geopolitically distant countries in shaping scientific fields through citation analysis, which may reveal how central these collaborations are in critical domains.

Geopolitical tensions may also be associated with reduced collaboration by limiting the movement of individuals. Scientific collaborations often originate from physical interactions between researchers, which can take many forms such as training, visits, and conferences. Measuring how barriers to travel relate to collaboration in modern contexts is therefore a promising direction for future research. Relatedly, the asymmetric associations we document suggest the possibility of genuine hysteresis in collaboration networks, analogous to the incumbency effects formalized in structural gravity models of trade (Egger et al., 2025); testing this with dynamic specifications that track whether collaboration recovers after geopolitical relations improve, or whether the damage persists, represents a promising direction for future work.

Country-specific studies are important as well. We find that geopolitical distance generally has a stronger association with collaboration when the United States and China are excluded from the analysis, suggesting that these countries are particularly resilient to geopolitical distance in overall patent collaboration. At the same time, the United States and China are key drivers of the differential sensitivity observed in critical technologies. These two findings are complementary: US–China dyads dampen the average patent response to geopolitical distance but account for most of the critical-technology interaction.

At the same time, knowledge diffusion and geopolitical tensions are influenced, and may to some extent be mitigated, by non-state actors. Saxenian (2006) and Qiao et al. (2024) highlight the importance of foreign knowledge workers acquiring knowledge in Western countries and subsequently transferring this knowledge back to their home countries upon return. Diffusion in this case occurs through the movement of individuals rather than through collaboration between them. A similar mechanism applies to multinational companies that operate across countries and are able to internally diffuse knowledge across locations without relying on external collaborations. Institutional proximity in this case (e.g., Boschma, 2005) may be less binding, as it can be offset by organizational proximity.

Although UN voting data offers a robust framework for analyzing geopolitical alignment, it may oversimplify the complexities of international relations. Future research could incorporate more granular measures of political and economic ties to provide a more comprehensive picture of collaboration dynamics.

Future research focusing on technological collaborations could also explore differences by type of applicant, distinguishing between corporate and academic collaborators. Although we provide results in which collaborators' locations are based on applicants' locations, a more detailed analysis, beyond the scope of this paper, is required to understand ownership strategies and the organizational structure of R&D activities. Our focus on inventors' locations is justified because our analysis aims to capture changes in collaborations between the places where knowledge is produced, rather than changes in firms' strategies, which often reflect multiple patent filing practices that are difficult to identify in a global context. Applicants' locations may instead reflect preferences for legal regimes, tax-related incentives, and profit-maximization considerations, particularly for multinational companies. More broadly, because our patent sample is restricted to foreign-oriented (international) patent families, observed declines in co-inventorship could partly reflect a reduced propensity to seek patent protection abroad rather than a genuine reduction in joint innovation. Country-year fixed effects absorb general shifts in filing propensity at the country level, and the robustness of our results to an applicant-based collaboration measure (Section 5.3) provides reassurance, but we cannot fully rule out this margin.

Finally, a promising avenue for future research is to more rigorously identify the causal impact of geopolitical tensions on international collaboration (and potentially vice versa). This would require

an identification strategy that generates plausibly exogenous variation in geopolitical relations (e.g., major events such as the sudden introduction of trade tariffs or military conflicts that upend global relations). Separately, assembling data on forms of collaboration not fully captured in scientific publications and patents – such as informal networks partnerships – would complement the formal collaboration measures used in this study. The evolution of the impact of geopolitical alignment on international collaboration networks over time is also relevant, particularly given the rise of China in science (Ramirez and Petersen, 2026) and how this has reshaped collaborative networks.

Ultimately, the resilience of the global R&I ecosystem amid growing fragmentation offers cautious optimism. It is imperative, however, to address these challenges proactively to ensure that international collaborations continue to support scientific and technological progress in tackling complex global challenges.

CRediT authorship contribution statement

Shreyas Gadgin Matha: Writing – review & editing, Writing – original draft, Methodology, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Matte Hartog:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Christian Chacua:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Pierre-Alexandre Balland:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Ricardo Hausmann:** Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Muhammed Ali Yildirim:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Conceptualization.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Sebastian Bustos, David Torun, other members of the Growth Lab, and participants at the EPIP 2025 conference for helpful comments and suggestions.

Appendix A. List of critical fields

See Table A.1.

Appendix B. Sample characteristics and descriptive statistics

Our empirical analysis relies on a comprehensive dataset that combines international collaboration data with measures of geopolitical alignment and standard gravity variables. Table 1 presents descriptive statistics for the main variables used in our gravity model analysis. The data reveal substantial variation in collaboration patterns across country pairs. Scientific collaborations show considerable heterogeneity, with a mean of 667 co-authored publications per country pair but a median of only 75, indicating a highly skewed distribution where most country pairs engage in relatively few collaborations while a small number of pairs account for the majority of international scientific partnerships.

The distribution of technological collaborations through patents is even more concentrated, with a mean of 11.9 collaborative patents

Table A.1

Critical fields.

Source: Own construction, based on [European Commission \(2022a,b\)](#) and [European Commission et al. \(2023\)](#).

No.	Critical technology field
1	Advanced connectivity and navigation technologies
2	Advanced materials, manufacturing, and recycling
3	Advanced sensing technologies
4	Advanced semiconductors technologies
5	Artificial Intelligence
6	Battery technology
7	Biofuels
8	Blockchain
9	Carbon capture and storage technologies
10	Cloud and edge computing
11	Cybersecurity
12	Geothermal energy
13	Grid technologies
14	Hydropower
15	Hydrogen
16	Industrial and environmental biotechnologies
17	Internet of Things
18	Medical and pharmaceutical biotechnologies
19	Nuclear energy
20	Quantum technologies
21	Robotics and autonomous systems
22	Solar energy
23	Space and propulsion technologies
24	Virtual reality and augmented reality
25	Wind energy

Notes: This list of critical technologies was curated and kindly provided by members of the team of the Chief Economist unit of DG Research and Innovation of the European Commission: Alexandr Hobza, Julien Ravet, Erik Canton, Valentina Di Girolamo & Florence Benoit.

per country pair and a median of zero. This stark difference between scientific and technological collaboration reflects the distinct nature of knowledge production in these domains. Scientific research benefits more readily from international partnerships, while technological development often remains more closely tied to domestic industrial capabilities and intellectual property considerations.

Our key explanatory variable, the ideal point distance measure based on UN voting patterns, shows a mean of 1.08 with substantial variation (standard deviation of 0.83). This measure captures geopolitical alignment, with higher values indicating greater geopolitical distance between countries. Notably, the three-year change in ideal point distance has a slightly negative mean (−0.07), suggesting a modest trend toward convergence in UN voting patterns over our sample period, though with considerable variation around this trend.

The categorical control variables in [Table B.2](#) reveal that approximately 44% of country pairs belong to different geopolitical blocs according to our UN voting-based classification, providing substantial variation for identifying the effects of geopolitical alignment. The correlation matrix in [Table 2](#) shows that scientific and technological collaborations are strongly positively correlated (0.853), while geopolitical distance measures show relatively low correlations with collaboration outcomes, suggesting that the relationships are subtle and require careful econometric analysis to detect.

Appendix C. Additional tables and figures

Interaction specification for directional asymmetry. As a robustness check for the decomposition specification in Eq. (6), we also estimate an interaction specification that includes the original (signed) IPD change interacted with a binary divergence indicator:

$$\mathbb{E}[Y_{ijt} | \cdot] = \exp\left(\beta_0 \Delta_3 \text{IPD}_{ijt} + \beta_1 \mathbf{1}[\Delta_3 \text{IPD}_{ijt} > 0]\right)$$

Table B.1

Data sources and coverage.

Source: Own construction.

Variable/Category	Data source and coverage
Geographic and cultural distances	CEPII GeoDist database (Mayer and Zignago, 2011) - common border, common language, common colonizer, common legal origins, geographic distance
Bilateral trade	Atlas of Economic Complexity (The Harvard Growth Lab, 2025) - 2000–2023
R&D expenditure	World Bank World Development Indicators (World Bank, 2024) (R&D as % of GDP) - 2000–2023
Human capital	World Bank World Development Indicators (World Bank, 2024) (share of labor force with advanced education, %) - 2000–2023
Critical technologies	European Commission (2022a,b) and European Commission et al. (2023) - retrospectively applied for 2000–2023
Ideal point estimates	UN General Assembly voting data (Bailey et al., 2017 ; Voeten et al., 2009) - dataset covers 1946–2023, 2000–2023 used in the paper
Scientific collaborations	OpenAlex data on scientific publications (Priem et al., 2022) - dataset goes back to the 19th century, 2000–2023 used in the paper
Technological collaborations	PATSTAT data on patents (European Patent Office, 2024) - dataset goes back to the 19th century, 2000–2023 used in the paper

Table B.2

Summary statistics: Categorical control variables.

Source: Own construction, using geopolitical alignment from UN General Assembly voting records, and borders and cultural distances from CEPII.

		Percent
Different Bloc	Same bloc	56.39
	Different bloc	43.61
Border sharing	No shared border	96.35
	Shared border	3.65
Language	Different language	89.81
	Common language	10.19
Colonial history	Different colonizer	95.22
	Common colonizer	4.78
Legal origins	Different legal origins	65.51
	Common legal origins	34.49

Notes: Different Bloc: Whether countries belong to different blocs in UN General Assembly. Border sharing: Whether countries share a land border. Language: Whether countries share a common official language. Colonial history: Whether countries had the same colonizer after 1945. Legal origins: Whether countries share common legal system origins.

$$+ \beta_2 \Delta_3 \text{IPD}_{ijt} \times \mathbf{1}[\Delta_3 \text{IPD}_{ijt} > 0] + \theta_{ij} + \gamma_{it} + \delta_{jt}). \quad (\text{C.1})$$

Here, β_0 captures the slope of IPD changes during convergence episodes ($\Delta_3 \text{IPD}_{ijt} \leq 0$), $\beta_0 + \beta_2$ gives the slope during divergence episodes ($\Delta_3 \text{IPD}_{ijt} > 0$), and β_2 directly measures the asymmetry—the additional slope when diverging relative to converging. A significant $\beta_2 < 0$ indicates that divergence episodes are associated with disproportionately larger reductions in collaboration. [Table C.8](#) reports results from both the decomposition and this interaction specification.

Data availability

Data will be made available on request.

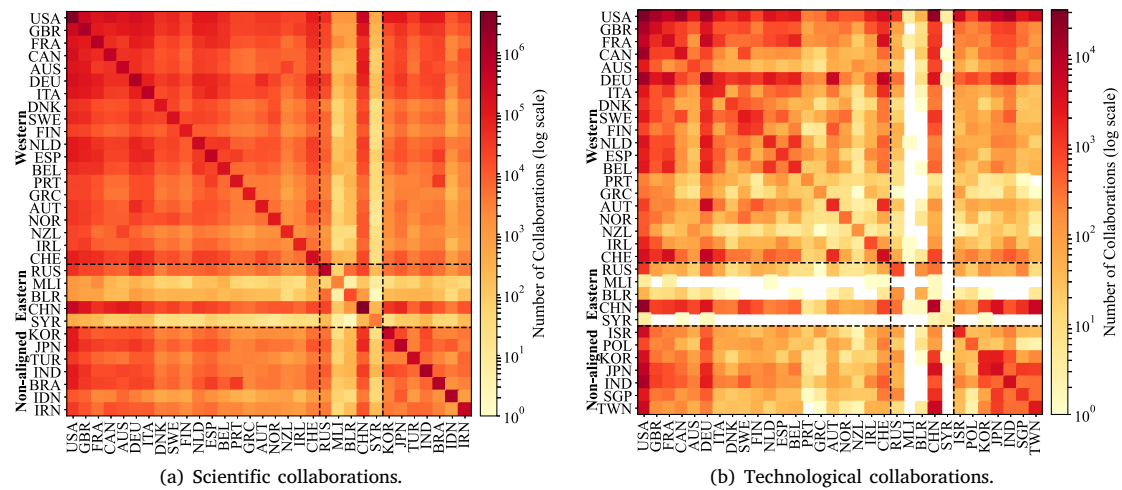


Fig. C.1. Heatmap of collaboration patterns - Counts.

Notes: Countries are ordered by geopolitical blocs and ideal point estimates by Bailey et al. (2017). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Own construction, using scientific publications from OpenAlex and patents from PATSTAT.

Table C.1

Geopolitical distance and international technological collaboration - Geopolitical blocs.

Source: Own construction using patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)
Different bloc $\times$ Critical technology			−0.026 (0.108)	−0.042 (0.047)
Different bloc	−0.035 (0.025)	−0.178*** (0.022)	−0.130** (0.050)	−0.159*** (0.021)
Critical technology			1.027*** (0.069)	
Geographic distance (log)	−0.407*** (0.011)	−0.095*** (0.018)	−0.077** (0.030)	−0.096*** (0.018)
Common border		−0.044 (0.030)	−0.045 (0.059)	−0.043 (0.031)
Common language		0.591*** (0.031)	0.471*** (0.050)	0.590*** (0.031)
Common colonizer		−0.428*** (0.058)	−0.350** (0.118)	−0.431*** (0.058)
Common legal origins		−0.026 (0.017)	−0.025 (0.037)	−0.025 (0.017)
Bilateral trade (log)		0.219*** (0.014)	0.161*** (0.026)	0.217*** (0.015)
Observations	1 960 628	1 958 156	1 958 156	1 958 156
Pseudo R ²	0.361	0.374	0.233	0.374
Squared Cor.	0.498	0.535	0.163	0.535
Country $i \times$ Year FE	X	X	X	X
Country $j \times$ Year FE	X	X	X	X
Technology FE	X	X		X

Notes: The table reports the results of PPML estimation, where the dependent variable is the number of collaborative patents in each IPC subclass. The sample spans 2000–2023. Standard errors (in parentheses) are clustered at the (country $i \times$ country $j \times$ technology) level.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.2

IPD changes and collaboration in patents by technology - Critical technologies.

Source: Own construction using patents from PATSTAT and geopolitical alignment from UN General Assembly voting records.

	(1)	(2)	(3)	(4)
IPD change (1 yr) × Critical technology		−0.092*** (0.017)		
IPD change (3 yr) × Critical technology			−0.148*** (0.021)	
IPD change (5 yr) × Critical technology				−0.209*** (0.026)
IPD change (1 year)		0.051*** (0.008)		
IPD change (3 years)	0.013+ (0.007)		0.075*** (0.010)	
IPD change (5 years)				0.102*** (0.013)
Observations	1 227 534	1 605 660	1 227 534	1 009 516
Pseudo R ²	0.439	0.416	0.439	0.453
Squared Cor.	0.651	0.632	0.652	0.662
Country i × Year FE	X	X	X	X
Country j × Year FE	X	X	X	X
Country-pair FE	X	X	X	X
Technology FE	X	X	X	X

Notes: PPML estimation. Dependent variable: collaborative patents in each IPC subclass. Sample spans 2000–2023. Standard errors clustered at (country i × country j × technology) level.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.3

IPD changes and collaboration in scientific publications.

Source: Own construction, using scientific publications from OpenAlex, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPIL.

	(1)	(2)	(3)	(4)	(5)
IPD change (1 yr)			−0.023*** (0.003)		
IPD change (3 yr)	−0.070*** (0.013)	−0.075*** (0.013)		−0.031*** (0.004)	
IPD change (5 yr)					−0.029*** (0.004)
Geographic distance (log)	−0.404*** (0.021)	−0.376*** (0.021)			
Common border		−0.109+ (0.056)			
Common language		0.382*** (0.064)			
Common colonizer		0.339** (0.119)			
Common legal origins		0.054+ (0.033)			
Observations	217 980	217 980	234 722	217 980	201 236
Pseudo R ²	0.952	0.956	0.994	0.994	0.994
Squared Cor.	0.908	0.898	0.997	0.997	0.997
Country i × Year FE	X	X	X	X	X
Country j × Year FE	X	X	X	X	X
Country-pair FE			X	X	X

Notes: Poisson pseudo-maximum likelihood estimation. Dependent variable: collaborative publications. Sample spans 2000–2023. Standard errors clustered at dyad level. Columns (1)–(2): Country × year FE. Columns (3)–(5): Country × year + pair FE.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.4

IPD changes and collaboration in patents.

Source: Own construction using patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)	(5)
IPD change (1 yr)			−0.019* (0.008)		
IPD change (3 yr)	−0.007 (0.017)	0.018 (0.016)		−0.023* (0.010)	
IPD change (5 yr)					−0.017 (0.012)
Geographic distance (log)	−0.649*** (0.030)	−0.471*** (0.030)			
Common border		0.183* (0.084)			
Common language		0.628*** (0.080)			
Common colonizer		−0.293+ (0.167)			
Common legal origins		0.126* (0.056)			
Observations	201 362	201 362	144 622	134 756	124 432
Pseudo R ²	0.936	0.945	0.964	0.964	0.964
Squared Cor.	0.921	0.953	0.995	0.995	0.995
Country i × Year FE	X	X	X	X	X
Country j × Year FE	X	X	X	X	X
Country-pair FE			X	X	X

Notes: Poisson pseudo-maximum likelihood estimation. Dependent variable: collaborative patents. Sample spans 2000–2023. Standard errors clustered at dyad level. Columns (1)–(2): Country × year FE. Columns (3)–(5): Country × year + pair FE.

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table C.5

IPD levels and international scientific collaboration.

Source: Own construction, using scientific publications from OpenAlex, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)
IPD × Different bloc				0.245*** (0.050)
IPD, lagged 1 yr		−0.086** (0.033)	−0.028 (0.031)	−0.162*** (0.034)
Different bloc	−0.183*** (0.027)			−0.478*** (0.060)
Geographic distance (log)	−0.410*** (0.024)	−0.388*** (0.021)	−0.211*** (0.024)	−0.168*** (0.023)
Common border			−0.145** (0.053)	−0.147** (0.050)
Common language			0.353*** (0.057)	0.347*** (0.057)
Common colonizer			0.377** (0.119)	0.247* (0.113)
Common legal origins			0.004 (0.032)	0.003 (0.031)
Bilateral trade (log)			0.182*** (0.018)	0.189*** (0.016)
Observations	264 590	234 904	229 218	229 218
Pseudo R ²	0.953	0.954	0.960	0.963
Squared Cor.	0.902	0.902	0.903	0.922
Country i × Year FE	X	X	X	X
Country j × Year FE	X	X	X	X

Notes: The table reports the results of Poisson pseudo-maximum likelihood estimation, where the dependent variable is the number of collaborative publications. The sample spans 2000–2023. Standard errors (in parentheses) are clustered at the dyad level.

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table C.6

IPD levels and international technological collaboration.

Source: Own construction using patents from PATSTAT, geopolitical alignment from UN General Assembly voting records, and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)
IPD × Different bloc				0.335*** (0.045)
IPD, lagged 1 yr		−0.002 (0.070)	0.096** (0.037)	−0.156*** (0.046)
Different bloc	0.137* (0.070)			−0.606*** (0.081)
Geographic distance (log)	−0.723*** (0.039)	−0.649*** (0.033)	−0.225*** (0.047)	−0.144*** (0.037)
Common border			0.095 (0.069)	0.132* (0.061)
Common language			0.668*** (0.073)	0.575*** (0.073)
Common colonizer			−0.208 (0.202)	−0.251 (0.195)
Common legal origins			0.037 (0.052)	0.075 (0.051)
Bilateral trade (log)			0.319*** (0.035)	0.320*** (0.029)
Observations	243 528	216 724	212 366	212 366
Pseudo R ²	0.933	0.936	0.948	0.950
Squared Cor.	0.917	0.922	0.964	0.972
Country i × Year FE	X	X	X	X
Country j × Year FE	X	X	X	X

Notes: The table reports the results of Poisson pseudo-maximum likelihood estimation, where the dependent variable is the number of collaborative patents. The sample spans 2000–2023. Standard errors (in parentheses) are clustered at the dyad level.

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table C.7

IPD levels and international technological collaboration in critical technologies.

Source: Own construction, using patents from PATSTAT and cultural and geographic distances from CEPII.

	(1)	(2)	(3)	(4)
IPD × Critical technology			0.198*** (0.038)	0.211*** (0.025)
IPD, lagged	−0.060** (0.021)	0.005 (0.017)	−0.089*** (0.026)	−0.093*** (0.014)
Critical technology			0.786*** (0.070)	
Geographic distance (log)	−0.394*** (0.010)	−0.075*** (0.014)	−0.043 (0.028)	−0.065*** (0.014)
Common border		0.036 (0.026)	0.004 (0.057)	0.016 (0.026)
Common language		0.508*** (0.030)	0.388*** (0.055)	0.510*** (0.029)
Common colonizer		−0.132+ (0.072)	−0.099 (0.140)	−0.134+ (0.075)
Common legal origins		−0.016 (0.019)	−0.027 (0.042)	−0.027 (0.018)
Bilateral trade (log)		0.254*** (0.013)	0.199*** (0.025)	0.266*** (0.013)
Observations	1 605 662	1 604 675	1 604 675	1 604 675
Pseudo R ²	0.380	0.392	0.237	0.395
Squared Cor.	0.549	0.586	0.174	0.606
Country i × Year FE	X	X	X	X
Country j × Year FE	X	X	X	X
Technology FE	X	X		X

Notes: The table reports the results of PPML estimation, where the dependent variable is the number of collaborative patents in each IPC subclass. The sample spans 2000–2023. Standard errors (in parentheses) are clustered at the (country i × country j × technology) level.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.8

Directional asymmetry: Decomposition and interaction specifications.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, and geopolitical alignment from UN General Assembly voting records.

	Publications			Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
IPD change (3 yr)	−0.031*** (0.004)		−0.026** (0.008)	−0.023* (0.010)		−0.021 (0.018)
IPD divergence (3 yr)		−0.047*** (0.009)			−0.008 (0.022)	
IPD convergence (3 yr)		0.017* (0.008)			0.036* (0.017)	
IPD change (3 yr) × Diverging			−0.030* (0.015)			0.031 (0.033)
Diverging indicator			0.005*** (0.001)			−0.007 (0.005)
Observations	217 980	217 980	217 980	134 756	134 756	134 756
Pseudo R ²	0.994	0.994	0.994	0.964	0.964	0.964
Squared Cor.	0.997	0.997	0.997	0.995	0.995	0.995
Country i × Year FE	X	X	X	X	X	X
Country j × Year FE	X	X	X	X	X	X
Country-pair FE	X	X	X	X	X	X
Symmetry test (χ^2)		3.88			0.71	
Symmetry test (p -value)		[0.049]			[0.400]	
$ \beta_{\text{div}} /\beta_{\text{conv}}$		2.79			0.22	

Notes: PPML estimates. Columns (1) and (4) replicate the symmetric specification; columns (2) and (5) report the decomposition into divergence and convergence; columns (3) and (6) interact the overall IPD change with a diverging indicator ($I[\Delta_3 \text{IPD} > 0]$). Standard errors (in parentheses) clustered at the dyad level. All models include country i × year, country j × year, and country-pair fixed effects.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.9

Directional asymmetry: Cross-scenario robustness.

Source: Own construction, using scientific publications from OpenAlex, patents from PATSTAT, and geopolitical alignment from UN General Assembly voting records.

Specification	Baseline	Q 10	Q 50	Applicant	No small countries	No CN/US
Publications						
Divergence (Decomp)	−0.047*** (0.009)	−0.031** (0.011)	−0.043*** (0.013)	−0.042*** (0.009)	−0.049*** (0.010)	−0.074*** (0.009)
Convergence (Decomp)	0.017* (0.008)	0.038** (0.012)	0.028* (0.013)	0.018* (0.008)	0.009 (0.009)	−0.019* (0.008)
IPD change (Interaction)	−0.026** (0.008)	−0.049*** (0.011)	−0.037** (0.013)	−0.026*** (0.007)	−0.017* (0.009)	0.012 (0.009)
IPD change × Diverging (Interaction)	−0.030* (0.015)	0.006 (0.020)	−0.016 (0.023)	−0.024 (0.015)	−0.040* (0.017)	−0.094*** (0.014)
Patents						
Divergence (Decomp)	−0.008 (0.022)	−0.004 (0.036)	−0.009 (0.025)	−0.044 (0.048)	−0.014 (0.025)	−0.095** (0.033)
Convergence (Decomp)	0.036* (0.017)	0.046 (0.031)	0.030 (0.019)	0.065* (0.029)	0.028 (0.018)	0.013 (0.028)
IPD change (Interaction)	−0.021 (0.018)	−0.042 (0.034)	−0.024 (0.021)	−0.027 (0.034)	−0.016 (0.020)	0.018 (0.031)
IPD change × Diverging (Interaction)	0.031 (0.033)	0.043 (0.054)	0.023 (0.038)	0.029 (0.060)	0.016 (0.038)	−0.078 (0.050)

Notes: Each cell reports the coefficient estimate with standard errors in parentheses, from the preferred PPML specification with country $i \times$ year, country $j \times$ year, and country-pair fixed effects. Scenarios are as defined in Table 8. “Decomp” refers to the divergence/convergence decomposition; “Interaction” refers to the IPD change $\times$ diverging indicator specification.

References

- Ackers, L., 2005. Moving people and knowledge: Scientific mobility in the European union. *Int. Migr.* 43 (5), 99–131.
- Adams, J.D., Black, G.C., Clemmons, J.R., Stephan, P.E., 2005. Scientific teams and institutional collaborations: Evidence from US universities, 1981–1999. *Res. Policy* 34 (3), 259–285.
- Aghion, P., Antonin, C., Paluskiewicz, L., Stromberg, D., Wargon, R., Westin, K., 2024. Does Chinese Research Hinge on US Co-Authors? Evidence from the China Initiative. CEP Discussion Paper No. 1936, London School of Economics.
- Airaud, F.S., de Soys, F., Richards, K., Santacreu, A.M., 2025. Fragmentation? Revisiting the Ideal Point Distance Measure of Geopolitical Distance. Board of Governors of the Federal Reserve System, FEDS Notes.
- Aiyar, S., Malacrino, D., Presbitero, A.F., 2024. Investing in friends: The role of geopolitical alignment in FDI flows. *Eur. J. Political Econ.* 83, 102508.
- Akbaritabar, A., Theile, T., Zagheni, E., 2024. Bilateral flows and rates of international migration of scholars for 210 countries for the period 1998–2020. *Sci. Data* 11 (1), 816.
- Alperin, J.P., Portenoy, J., Demes, K., Larivière, V., Haustein, S., 2024. An analysis of the suitability of OpenAlex for bibliometric analyses. *arXiv preprint arXiv: 2404.17663*.
- Ascani, A., Bettarelli, L., Resmini, L., Bolland, P.-A., 2020. Global networks, local specialisation and regional patterns of innovation. *Res. Policy* 49 (8), 104031.
- Bahar, D., Choudhury, P., Miguelez, E., Signorelli, S., 2024. Global mobile inventors. *J. Dev. Econ.* 171, 103357.
- Bailey, M.A., Strezhnev, A., Voeten, E., 2017. Estimating dynamic state preferences from United Nations voting data. *J. Confl. Resolut.* 61 (2), 430–456.
- Baldwin, R., 2016. *The Great Convergence: Information Technology and the New Globalization*. Harvard University Press.
- Baldwin, R., Krugman, P., 1989. Persistent trade effects of large exchange rate shocks. *Q. J. Econ.* 104 (4), 635–654.
- Ball, M.M., 1951. Bloc voting in the general assembly. *Int. Organ.* 5 (1), 3–31.
- Bolland, P.-A., Boschma, R., Ravet, J., 2019. Network dynamics in collaborative research in the EU, 2003–2017. *Eur. Plan. Stud.* 27 (9), 1811–1837.
- Belenzon, S., Schankerman, M., 2013. Spreading the word: Geography, policy, and knowledge spillovers. *Rev. Econ. Stat.* 95 (3), 884–903.
- Boschma, R., 2005. Proximity and innovation: A critical assessment. *Reg. Stud.* 39 (1), 61–74.
- Bosone, C., Stamato, G., 2024. Beyond Borders: How Geopolitics Is Reshaping Trade. ECB Working Paper No. 2960.
- Breschi, S., Lissoni, F., 2001. Knowledge spillovers and local innovation systems: A critical survey. *Ind. Corp. Chang.* 10 (4), 975–1005.
- Castellani, D., Jimenez, A., Zanfei, A., 2013. How remote are R&D labs? Distance factors and international innovative activities. *J. Int. Bus. Stud.* 44 (7), 649–675.
- Chaney, T., 2018. The gravity equation in international trade: An explanation. *J. Political Econ.* 126 (1), 150–177.
- Charpin, R., London, J., Vincent, N., 2024. The effect of geopolitical tensions on international research collaborations and its implications for global operations management. *Int. J. Prod. Econ.* 268, 109120.
- Crescenzi, R., Dyèvre, A., Neffke, F., 2022. Innovation catalysts: How multinationals reshape the global geography of innovation. *Econ. Geogr.* 98 (3), 199–227.
- Dixit, A., 1989. Entry and exit decisions under uncertainty. *J. Political Econ.* 97 (3), 620–638.
- Egger, P., Foellmi, R., Schetter, U., Torun, D., 2025. Gravity with history: On incumbency effects in international trade. *J. Eur. Econ. Assoc.* 23 (4), 1350–1396.
- European Commission, 2022a. ERA Industrial Technology Roadmap for Low-Carbon Technologies in Energy-Intensive Industries. Technical report, Publications Office of the European Union, <https://data.europa.eu/doi/10.2777/92567>.
- European Commission, 2022b. Science, Research and Innovation performance of the EU, 2022. Technical report, SRIP, Publications Office of the European Union, <https://data.europa.eu/doi/10.2777/38888>.
- European Commission, Stres, S., Golebiowska-Tataj, D., Walz, R., et al., 2023. Research and Innovation to Thrive in the Poly-Crisis Age. Technical report, Directorate-General for Research and Innovation, Publications Office of the European Union, <https://data.europa.eu/doi/10.2777/92915>.
- European Patent Office, 2024. PATSTAT global 2024 spring edition.
- Feenstra, R.C., 2015. *Advanced International Trade: Theory and Evidence*. Princeton University Press.
- Flynn, R., Glennon, B., Murciano-Goroff, R., Xiao, J., 2024. Building a Wall Around Science: The Effect of U.S.-China Tensions on International Scientific Research. NBER Working Paper No. 32622.
- Gartzke, E., 2000. Preferences and the democratic peace. *Int. Stud. Q.* 44 (2), 191–212.
- Gopinath, G., Gourinchas, P.-O., Presbitero, A., Topalova, P.B., 2024. Changing Global Linkages: A New Cold War? Working Paper No. 2024/076, International Monetary Fund.
- Grabher, G., 1993. The weakness of strong ties: the lock-in of regional development in the Ruhr Area. In: *The Embedded Firm: On the Socioeconomics of Industrial Networks*. Routledge, London, pp. 255–277.
- Griffith, R., Lee, S., Van Reenen, J., 2011. Is distance dying at last? Falling home bias in fixed-effects models of patent citations. *Quant. Econ.* 2 (2), 211–249.
- Head, K., Li, Y.A., Minondo, A., 2019. Geography, ties, and knowledge flows: Evidence from citations in mathematics. *Rev. Econ. Stat.* 101 (4), 713–727.
- Head, K., Mayer, T., 2014. Gravity equations: Workhorse, toolkit, and cookbook. In: *Handbook of International Economics*. Vol. 4, Elsevier, pp. 131–195.
- Hou, L., Pan, Y., Zhu, J.J., 2021. Impact of scientific, economic, geopolitical, and cultural factors on international research collaboration. *J. Inf.* 15 (3), 101194.
- Jaffe, A.B., Trajtenberg, M., Henderson, R., 1993. Geographic localization of knowledge spillovers as evidenced by patent citations. *Q. J. Econ.* 108 (3), 577–598.
- Javorcik, B., Kitzmueller, L., Schweiger, H., Yıldırım, M.A., 2024. Economic costs of friendshoring. *World Econ.* 47 (7), 2871–2908.
- Kalemli-Özcan, Ş., Kwak, J.H., 2020. Capital flows and leverage. *Annu. Rev. Econ.* 12 (1), 833–846.
- Keller, W., Yeaple, S.R., 2013. The gravity of knowledge. *Am. Econ. Rev.* 103 (4), 1414–1444.
- Kerr, S.P., Kerr, W.R., 2018. Global collaborative patents. *Econ. J.* 128 (612), F235–F272.
- Kolb, M., 2018. What Is Globalization? And How Has the Global Economy Shaped the United States. Peterson Institute for International Economics, URL <https://www.piie.com/microsites/globalization/what-is-globalization>.

- Koremenos, B., 2005. Contracting around international uncertainty. *Am. Political Sci. Rev.* 99 (4), 549–565.
- Lee, J.J., Haupt, J.P., 2021. Scientific collaboration on COVID-19 amidst geopolitical tensions between the US and China. *J. High. Educ.* 92 (2), 303–329.
- Li, Y.A., 2014. Borders and distance in knowledge spillovers: Dying over time or dying with age?—Evidence from patent citations. *Eur. Econ. Rev.* 71, 152–172.
- Liebersohn, S., 1985. *Making It Count: The Improvement of Social Research and Theory*. University of California Press.
- Makkonen, T., Mitze, T., 2023. Geopolitical conflicts, sanctions and international knowledge flows: EU–Russia collaboration during the Ukraine crisis. *World Econ.* 46 (10), 2926–2949.
- Mayer, T., Zignago, S., 2011. Notes on CEPII's Distances Measures: The GeoDist Database. Working Paper 2011–25, CEPII, URL <http://www.cepii.fr/CEPII/en/publications/wp/abstract.asp?NoDoc=3877>.
- McCann, P., 2008. Globalization and economic geography: the world is curved, not flat. *Camb. J. Reg. Econ. Soc.* 1 (3), 351–370.
- Migueluez, E., Raffo, J., Chacua, C., Coda-Zabetta, M., Yin, D., Lissoni, F., Tarasconi, G., 2019. Tied In: The Global Network of Local Innovation. WIPO Economic Research Working Paper, 58.
- Montobbio, F., Sterzi, V., 2013. The globalization of technology in emerging markets: A gravity model on the determinants of international patent collaborations. *World Dev.* 44, 281–299.
- Moon, B.E., 1985. Consensus or compliance? Foreign-policy change and external dependence. *Int. Organ.* 39 (2), 297–329.
- Morrison, A., 2008. Gatekeepers of knowledge within industrial districts: Who they are, how they interact. *Reg. Stud.* 42 (6), 817–835.
- Nature Editorial, 2024. To boost science, the growing BRICS group must embrace inclusion and transparency. *Nature* 635, 258, Editorial.
- Patterson, N.K., Montanjes, M., Cardillo, C., Motala, J., 2004. Foreign Direct Investment: Trends, Data Availability, Concepts, and Recording Practices. International Monetary Fund.
- Picci, L., 2010. The internationalization of inventive activity: A gravity model using patent data. *Res. Policy* 39 (8), 1070–1081.
- Piermartini, R., Yotov, Y.V., 2016. Estimating Trade Policy Effects with Structural Gravity. Working Paper Series 6009, CESifo.
- Priem, J., Piwowar, H., Orr, R., 2022. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*.
- Qiao, Y., Ascani, A., Breschi, S., Morrison, A., 2024. Returnees and innovation: Evidence from Chinese publicly listed firms. *Sci. Public Policy* 51 (5), 861–878.
- Ramirez, A.M., Petersen, A.M., 2026. Transformation of Global Science core-periphery structure towards a multi-polar horizon: The rise of China and the Global South from 1980–2020. *Res. Policy* 55 (1), 105370.
- Ribeiro, L.C., Rapini, M.S., Silva, L.A., Albuquerque, E.M., 2018. Growth patterns of the network of international collaboration in science. *Scientometrics* 114, 159–179.
- Rovito, S., Kaushik, D., Aggarwal, S.D., 2021. The impact of international scientists, engineers, and students on US research outputs and global competitiveness. *MIT Sci. Policy Rev.* 2, 15–25.
- Santos Silva, J.M.C., Tenreiro, S., 2006. The log of gravity. *Rev. Econ. Stat.* 88 (4), 641–658.
- Saxenian, A., 2006. *The New Argonauts: Regional Advantage in a Global Economy*. Harvard University Press.
- Shih, T., Chubb, A., Cooney-O'Donoghue, D., 2024. Scientific collaboration amid geopolitical tensions: A comparison of Sweden and Australia. *High. Educ.* 87 (5), 1339–1356.
- Simard, M.-A., Basson, I., Hare, M., Larivière, V., Mongeon, P., 2024. The open access coverage of OpenAlex, Scopus and Web of Science. *arXiv preprint arXiv:2404.01985*.
- Storper, M., Venables, A.J., 2004. Buzz: Face-to-face contact and the urban economy. *J. Econ. Geogr.* 4 (4), 351–370.
- The Harvard Growth Lab, 2025. Bilateral trade data aggregated by year. URL <https://doi.org/10.7910/DVN/5NGVOB>, Version 1.1, Harvard Dataverse.
- The World Bank, 2023. *World Development Report 2023: Migrants, Refugees, and Societies*. The World Bank.
- Tinbergen, J., 1962. *Shaping the World Economy: Suggestions for an International Economic Policy*. Twentieth Century Fund, New York.
- Turkina, E., Van Assche, A., 2018. Global connectedness and local innovation in industrial clusters. *J. Int. Bus. Stud.* 49 (6), 706–728.
- Van Noorden, R., 2012. Global mobility: science on the move. *Nature* 490 (7420), 326–329.
- Velez-Estevez, A., Perez, I.J., García-Sánchez, P., Moral-Muñoz, J.A., Cobo, M.J., 2023. New trends in bibliometric APIs: A comparative analysis. *Inf. Process. Manage.* 60 (4), 103385.
- Veuglers, R., Cincera, M., Frietsch, R., Rammer, C., Schubert, T., Pelle, A., Renda, A., Montalvo, C., Leijten, J., 2015. The impact of Horizon 2020 on innovation in Europe. *Intereconomics* 50 (1), 4–30.
- Voeten, E., Strezhnev, A., Bailey, M., 2009. United Nations general assembly voting data. URL <https://doi.org/10.7910/DVN/LEJUQZ>, Harvard Dataverse, V33.
- Wagner, C.S., Jonkers, K., 2017. Open countries have strong science. *Nature* 550 (7674), 32–33.
- World Bank, 2024. *World Development Indicators*. The World Bank Group, URL <https://databank.worldbank.org/source/world-development-indicators>, (Accessed: 2024).
- York, R., Light, R., 2017. Directional asymmetry in sociological analyses. *Socius* 3, 1–13.
- Yotov, Y.V., Piermartini, R., Monteiro, J.-A., Larch, M., 2016. *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model*. WTO, URL https://www.wto-ilibrary.org/economic-research-and-trade-policy-analysis/advanced-guide-to-trade-policy-analysis-the-structural-gravity-model_abc0167e-en.
- Zhang, L., Cao, Z., Sivertsen, G., Kochetkov, D., 2024. The influence of geopolitics on research activity and international collaboration in science: The case of Russia. *Scientometrics* 129 (10), 6007–6021.